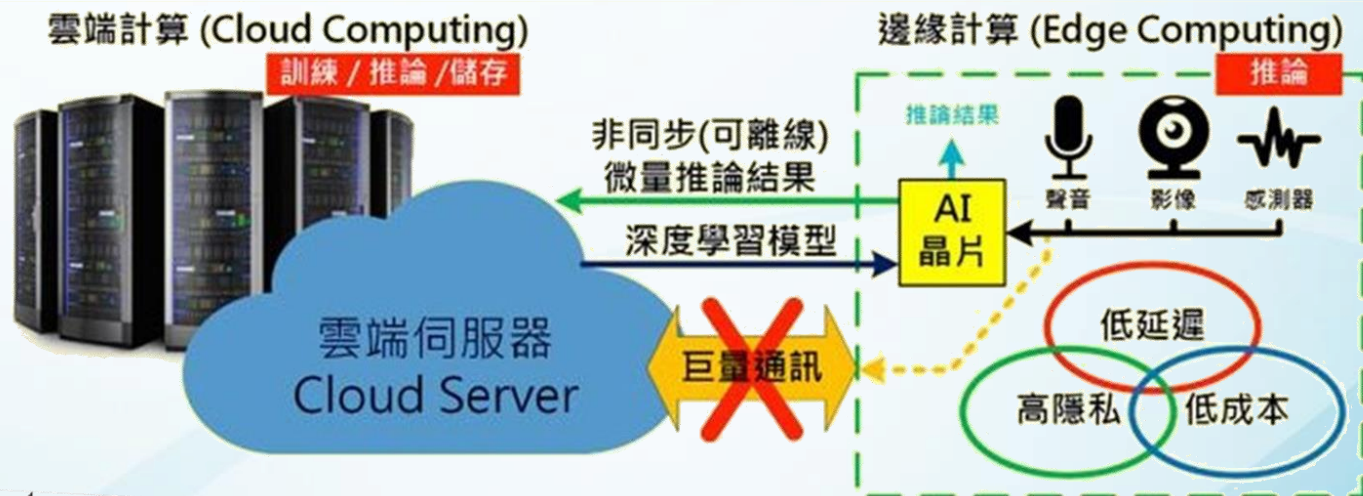
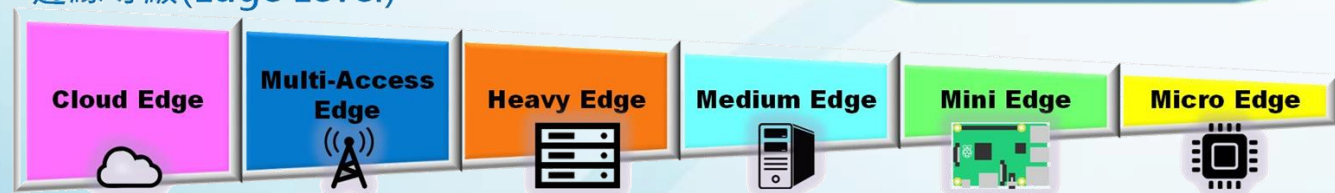


邊緣人工智慧實務 (EE5354701)



Edge AI

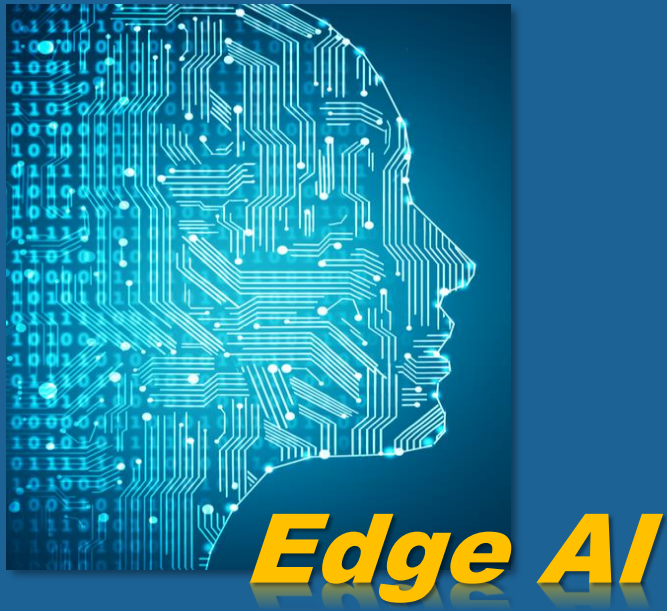
邊緣等級(Edge Level)



2. 邊緣運算硬體架構

電機工程系 人工智慧應用產碩專班 許哲豪 兼任助理教授

簡報大綱

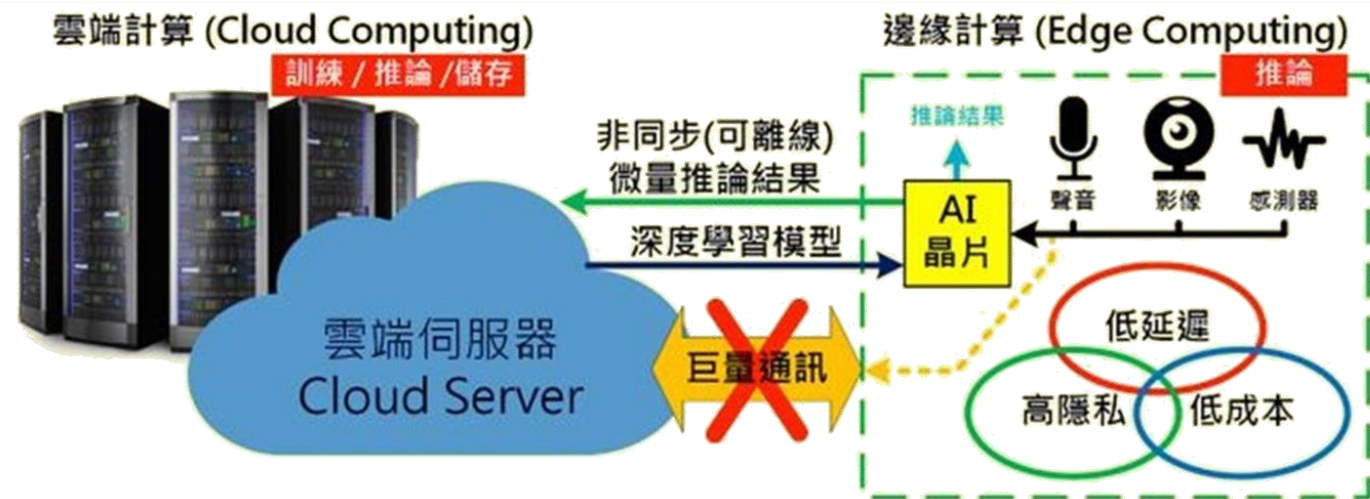


- 2.1. 基本運算原理
- 2.2. 加速運算晶片
- 2.3. 嵌入式級硬體
- 2.4. AI PC 級硬體

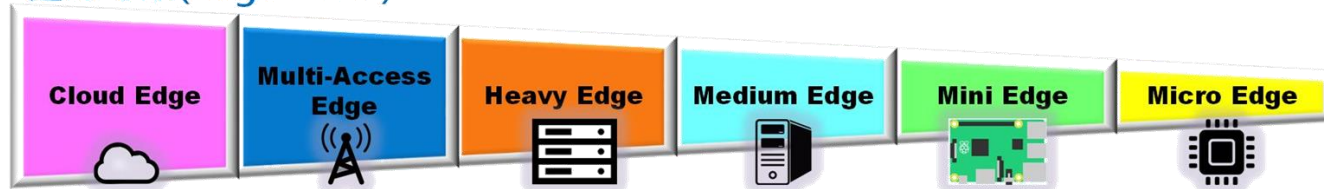
https://hackmd.io/@OmniXRI-Jack/NTUST_EdgeAI_2026



Edge AI

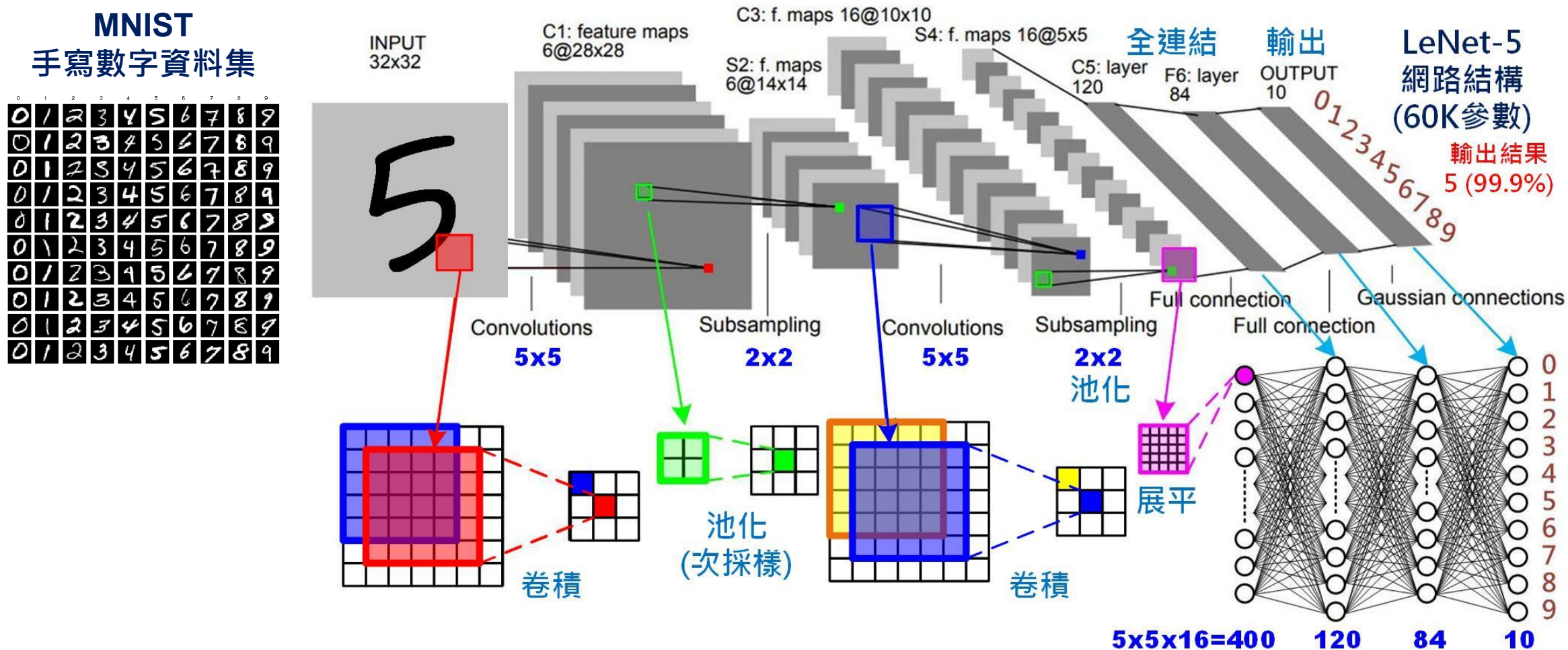


邊緣等級(Edge Level)



2.1. 基本運算原理

卷積神經網路(CNN)



資料來源：<http://omnixri.blogspot.com/2018/06/blog-post.html>

卷積神經網路 — 卷積

1	0	0	1	0
0	1	0	0	1
1	0	1	0	0
0	1	0	1	0
1	0	1	0	0

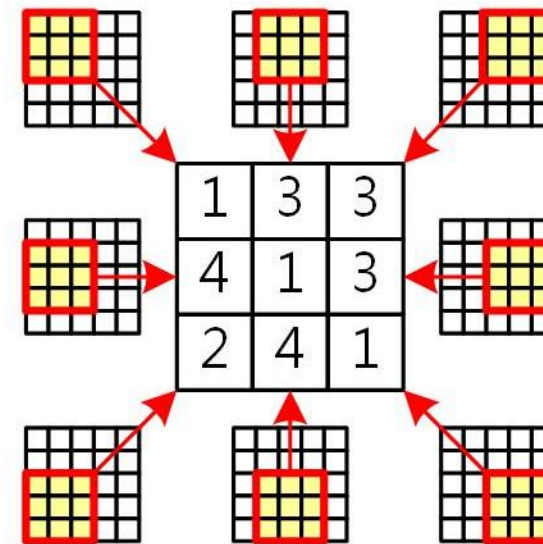
輸入矩陣

*

0	1	1
1	0	1
1	1	0

卷積核
(Kernel)

=



1	2	3
4	5	6
7	8	9

*

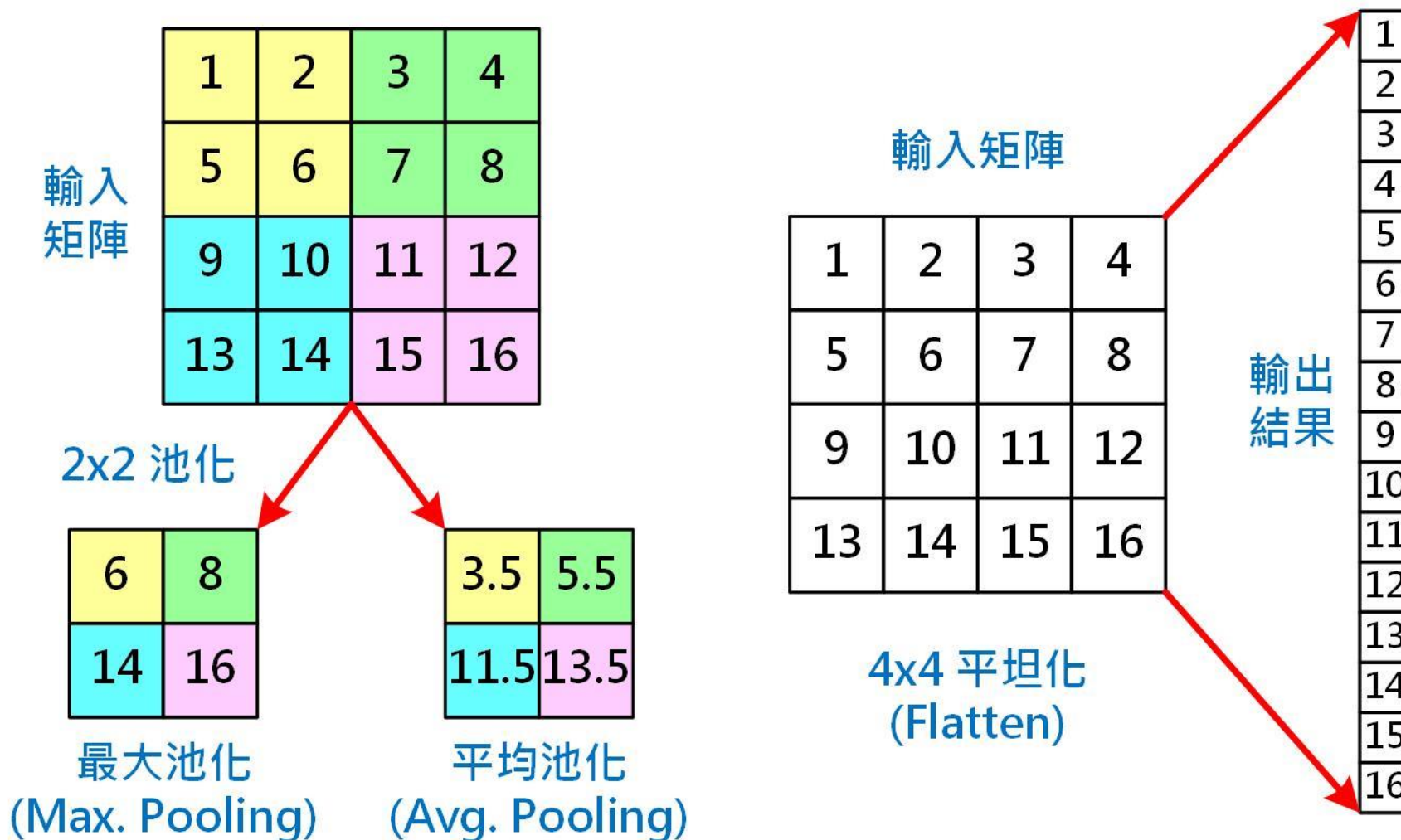
A	B	C
D	E	F
G	H	I

=

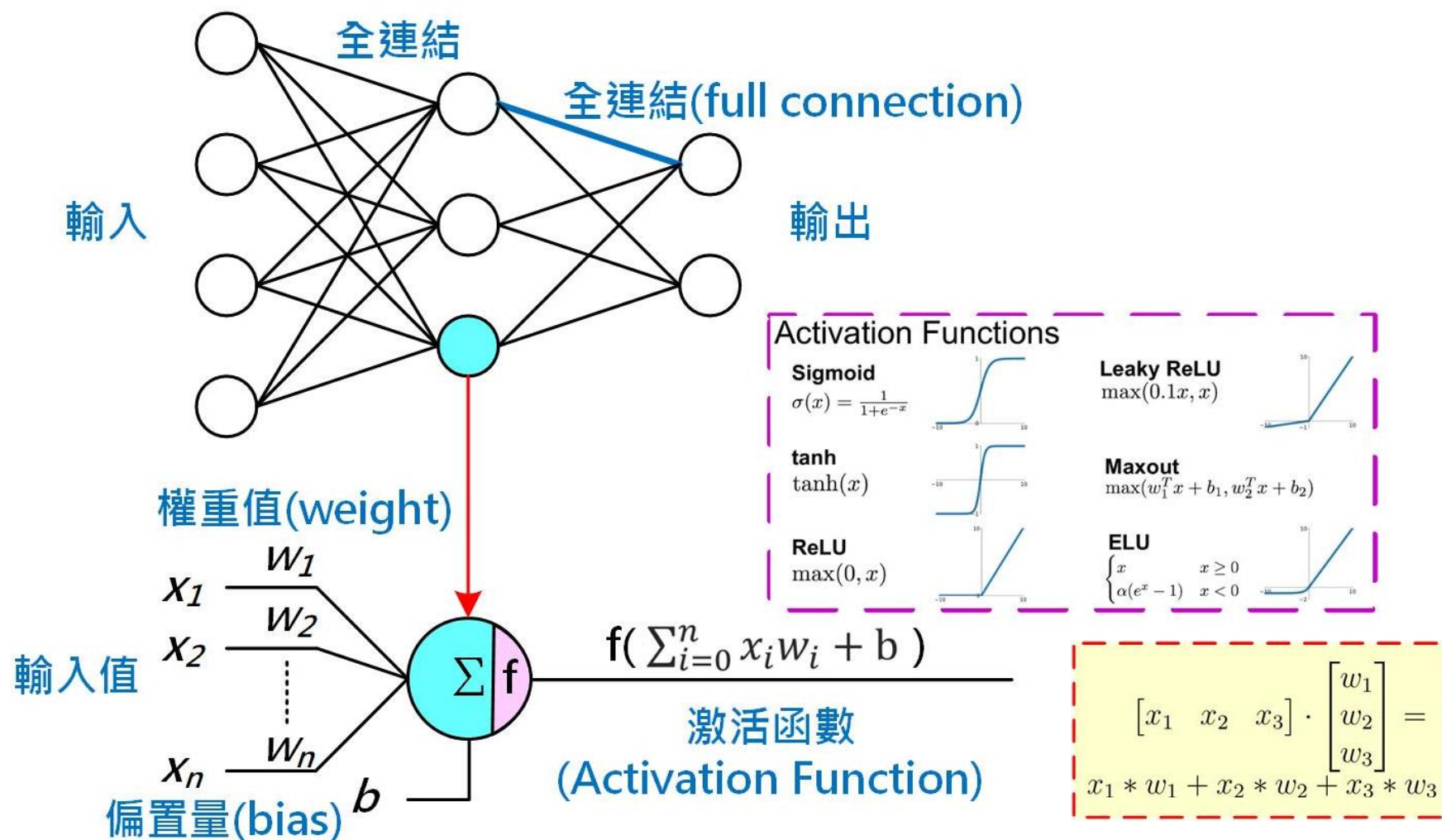
$$\begin{aligned}
 &1*A + 2*B + 3*C + \\
 &4*D + 5*E + 6*F + \\
 &7*G + 8*H + 9*I
 \end{aligned}$$

* 相當於Python numpy 的矩陣各元素相乘 $a*b$ 而非矩陣點乘 $a.dot(b)$

卷積神經網路 — 池化、平坦化



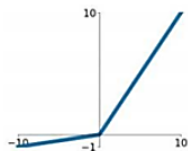
卷積神經網路 — 全連結



* 相當於Python numpy 的矩陣點乘 **a.dot(b)** 而非矩陣各元素相乘的 **a*b**

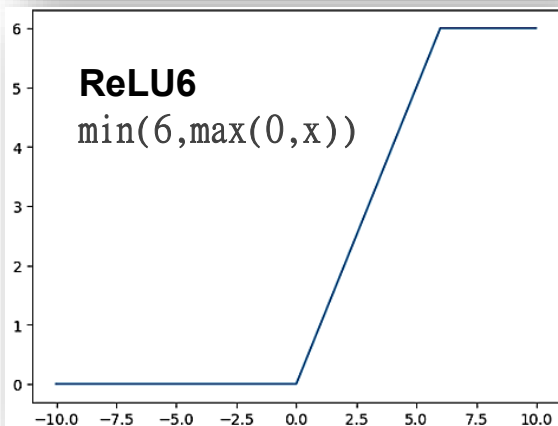
卷積神經網路 — 激活函數(1/2)

Leaky ReLU
 $\max(0.1x, x)$



ReLU6

$\min(6, \max(0, x))$



ReLU

$\max(0, x)$

GELU

$\frac{x}{2} \left(1 + \tanh \left(\sqrt{\frac{2}{\pi}} (x + ax^3) \right) \right)$

PReLU

$\max(\alpha x, x)$

ELU

$\begin{cases} x & \text{if } x > 0 \\ \alpha(x \exp x - 1) & \text{if } x < 0 \end{cases}$

Swish

$\frac{x}{1 + \exp -x}$

SELU

$\alpha(\max(0, x) + \min(0, \beta(\exp x - 1)))$

SoftPlus

$\frac{1}{\beta} \log(1 + \exp(\beta x))$

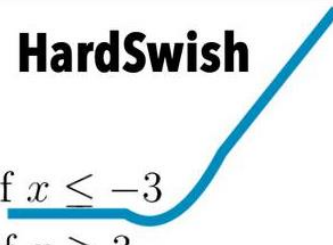
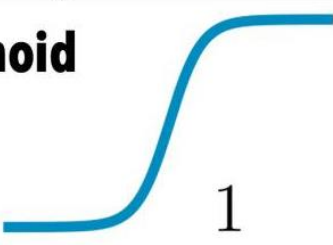
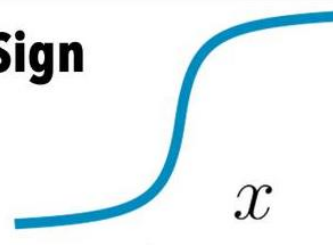
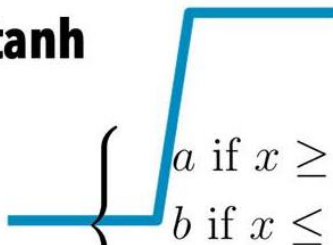
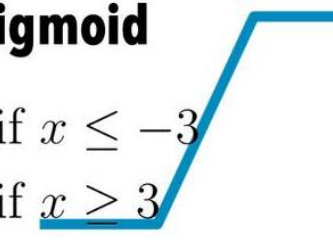
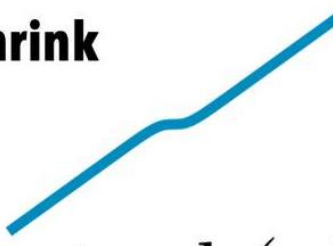
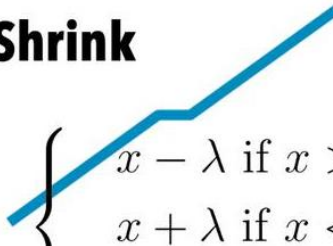
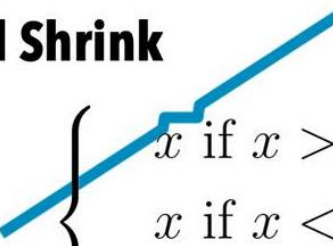
Mish

$x \tanh \left(\frac{1}{\beta} \log(1 + \exp(\beta x)) \right)$

RReLU

$\begin{cases} x & \text{if } x \geq 0 \\ ax & \text{if } x < 0 \text{ with } a \sim \mathcal{R}(l, u) \end{cases}$

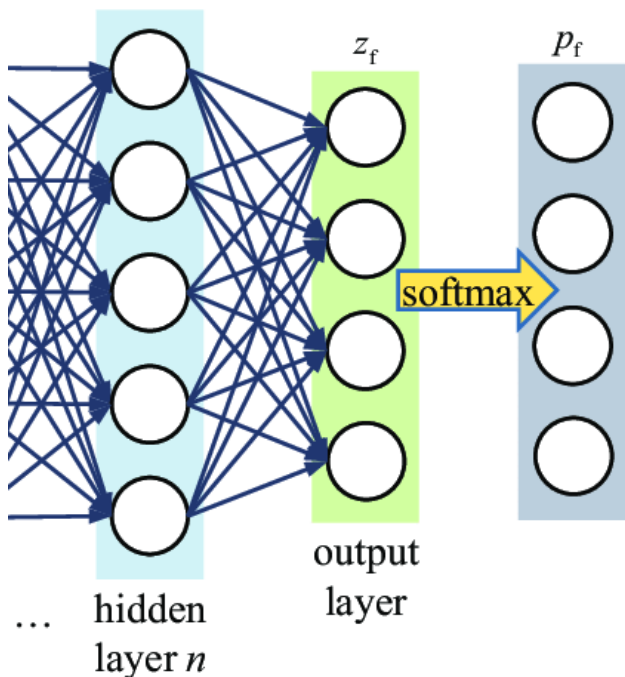
卷積神經網路 – 激活函數(2/2)

HardSwish  $\begin{cases} 0 & \text{if } x \leq -3 \\ x & \text{if } x \geq 3 \\ x(x+3)/6 & \text{otherwise} \end{cases}$	Sigmoid  $\frac{1}{1 + \exp(-x)}$	SoftSign  $\frac{x}{1 + x }$
Tanh  $\tanh(x)$	Hard tanh  $\begin{cases} a & \text{if } x \geq a \\ b & \text{if } x \leq b \\ x & \text{otherwise} \end{cases}$	Hard Sigmoid  $\begin{cases} 0 & \text{if } x \leq -3 \\ 1 & \text{if } x \geq 3 \\ x/6 + 1/2 & \text{otherwise} \end{cases}$
Tanh Shrink  $x - \tanh(x)$	Soft Shrink  $\begin{cases} x - \lambda & \text{if } x > \lambda \\ x + \lambda & \text{if } x < -\lambda \\ 0 & \text{otherwise} \end{cases}$	Hard Shrink  $\begin{cases} x & \text{if } x > \lambda \\ x & \text{if } x < -\lambda \\ 0 & \text{otherwise} \end{cases}$

卷積神經網路 — 輸出函數

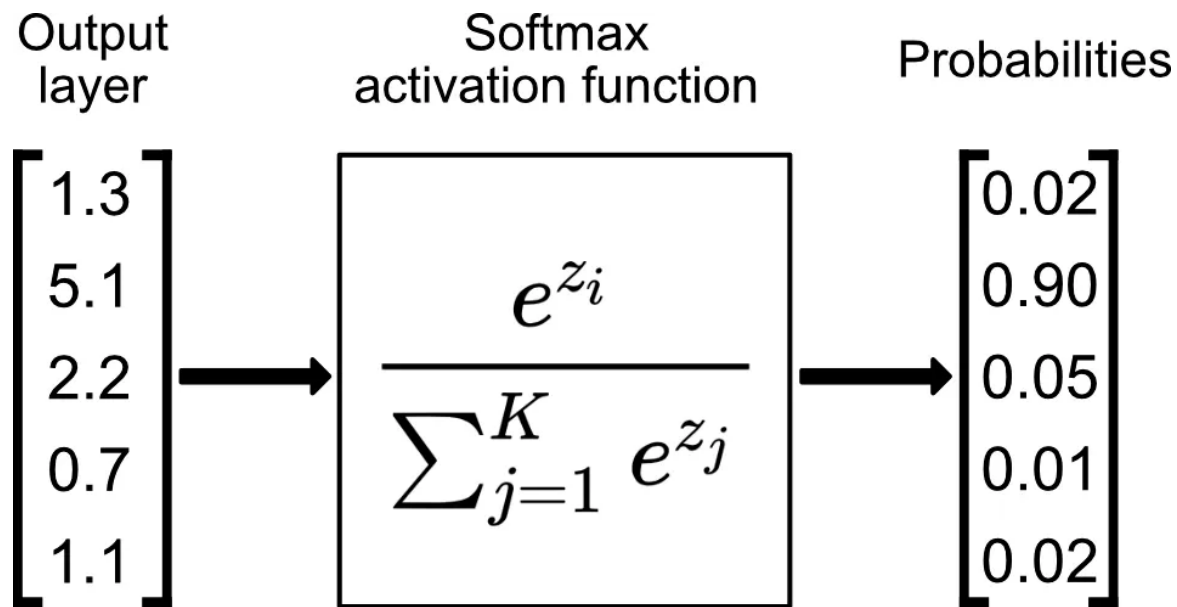
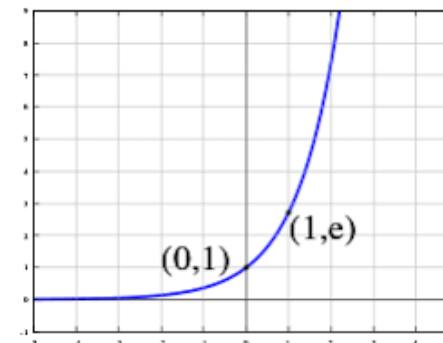
softmax()

歸一化指數函數
將所有發生機率
壓縮到0~1之間



exp()

以自然常數e為底
的指數函數



卷積神經網路 — 權重與計算量

Layer Name	Input W×H×D	Kernel W×H×D/S	Output W×H×D	Params	Mults
C1: conv2d	32×32×1	5×5×6	28×28×6	$1 \times 5 \times 5 \times 6 + 6 = 156$	$28 \times 28 \times 1 \times 5 \times 5 \times 6 = 117,600$
S2: pool/2	28×28×6	2×2/2	14×14×6	0	0
C3: conv2d	14×14×6	5×5×16	10×10×16	$6 \times 5 \times 5 \times 16 + 16 = 2,416$	$10 \times 10 \times 6 \times 5 \times 5 \times 16 = 240,000$
S4: pool/2	10×10×16	2×2/2	5×5×16	0	0
C5: conv2d	5×5×16	5×5×120	1×1×120	$16 \times 5 \times 5 \times 120 + 120 = 48,120$	$1 \times 1 \times 16 \times 5 \times 5 \times 120 = 48,000$
F6: conv2d	1×1×120	1×1×84	1×1×84	$120 \times 1 \times 1 \times 84 + 84 = 10,164$	$120 \times 84 = 10,080$
F7: conv2d	1×1×84	1×1×10	1×1×10	$84 \times 1 \times 1 \times 10 + 10 = 850$	$84 \times 40 = 840$
Total				61,706	416,520

LeNet-5 模型結構

圖片來源：

https://www.researchgate.net/publication/329970273_Image_recognition_by_knowledge_transfer_using_Deep_Convolutional_Neural_Network

數字表示系統 — 進制轉換

數字表示方式

	10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0
10進制 (Dec)	0	0	0	0	0	2	3	6
	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
2進制 (Bin)	1	1	1	0	1	1	0	0
	16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0
16進制 (Hex)	0	0	0	0	0	0	E	C

不同進制轉成10進制

$$2 \times 10^2 + 3 \times 10^1 + 6 \times 10^0 = 236 \text{ (Dec)}$$

$$1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 = 236 \text{ (Dec)}$$

$$14 \times 16^1 + 12 \times 16^0 = 236 \text{ (Dec)}$$

10進制轉成2進制

短除法

2	236	0	LSB
2	118	0	
2	59	1	餘數
2	29	1	
2	14	0	
2	7	1	
2	3	1	
2	1	1	MSB
	0	商數	

數字表示系統 — 整數

整數常用名稱及表示範圍

生成式AI用

邊緣推論用

AI用名	C/C++名稱	Byte	數值範圍
INT4	char	1	-8 ~ +7
	unsigned char	1	0 ~ +15
INT8	char	1	-128 ~ +127
	unsigned char	1	0 ~ +255
INT16	Short (Short int)	2	-32,768 ~ +32,767
	unsigned short	2	0 ~ +65,535
INT32	Int (long int)	4	-2,147,483,648 ~ +2,147,483,647
	unsigned int	4	0 ~ +4,294,967,295
INT64	long long	8	-9,223,372,036,854,775,808 ~ +9,223,372,036,854,775,807
	unsigned long long	8	0 ~ +18,446,744,073,709,551,615

負 (整) 數表示法

最高位元

1的補數

BIN	DEC	BIN	DEC
00000010	2	00000010	2
00000001	1	00000001	1
00000000	0	00000000	0
10000001	-1	11111110	-1
10000010	-2	11111101	-2

2的補數 (1的補數加1)

BIN	DEC
00000010	2
00000001	1
00000000	0
11111111	-1
11111110	-2

影像來源：<https://omnixri.blogspot.com/2023/02/vmaker-edge-ai-02-ai.html>

數字表示系統 — 浮點數

二進制定點數表示法



二進制浮點數表示法 (IEEE 754)

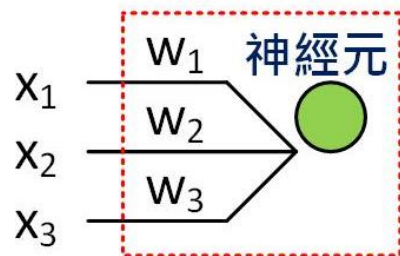
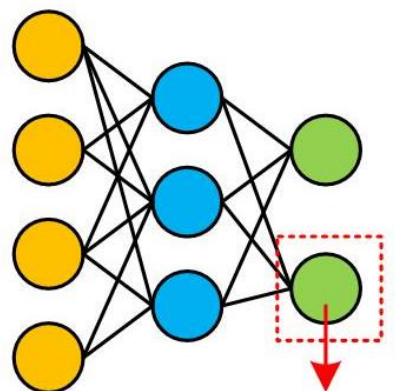


OmniXRI整理製作, 2023/2/13

影像來源: <https://omnixri.blogspot.com/2023/02/vmaker-edge-ai-02-ai.html>

數字表示系統 — AI 常用浮點數

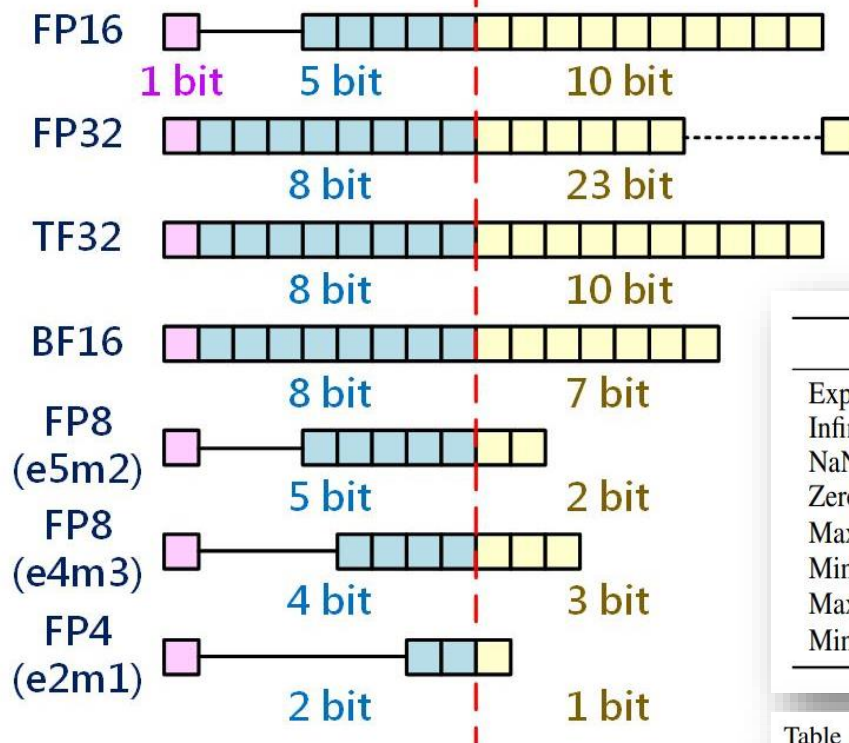
神經網路



$$y = x_1 * w_1 + x_2 * w_2 + x_3 * w_3$$

AI常用二進制浮點數表示法

符號 指數(exponent) 尾數(mantissa)



	E4M3 推理用	E5M2 訓練用
Exponent bias	7	15
Infinities	N/A	$S.11111.00_2$
NaN	$S.1111.111_2$	$S.11111.\{01, 10, 11\}_2$
Zeros	$S.0000.000_2$	$S.00000.00_2$
Max normal	$S.1111.110_2 = 1.75 * 2^8 = 448$	$S.11110.11_2 = 1.75 * 2^{15} = 57,344$
Min normal	$S.0001.000_2 = 2^{-6}$	$S.00001.00_2 = 2^{-14}$
Max subnorm	$S.0000.111_2 = 0.875 * 2^{-6}$	$S.00000.11_2 = 0.75 * 2^{-14}$
Min subnorm	$S.0000.001_2 = 2^{-9}$	$S.00000.01_2 = 2^{-16}$

Table 1: Numbers represented in FP4-E2M1 with NaN and Inf (IEEE 754 standard) and Numbers represented in FP4-E2M1 without NaN and Inf (Our design).

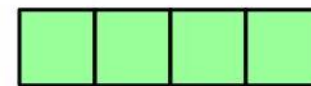
UINT4	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
FP4 w/ NaN&Inf	0	0.5	1	1.5	2	3	Inf	NaN	-0	-0.5	-1	-1.5	-2	-3	Inf	NaN
FP4 w/o NaN&Inf	0	0.5	1	1.5	2	3	4	6	-0	-0.5	-1	-1.5	-2	-3	-4	-6

影像來源：
<https://omnixri.blogspot.com/2023/02/vmaker-edge-ai-02-ai.html>
<https://omnixri.blogspot.com/2024/03/nvidia-gtc-2024-fp8fp4-ai.html>

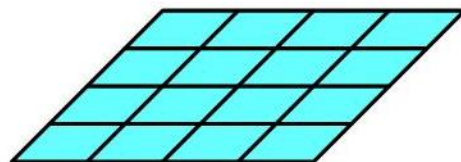
資料維度



0維 純量
(scalar)

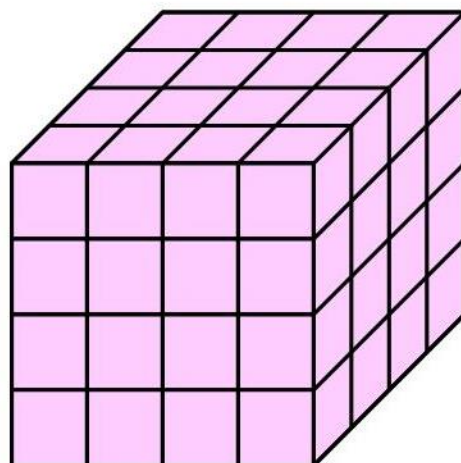
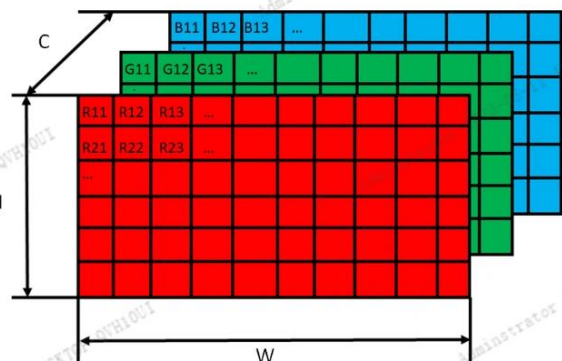


1維 向量
(vector)

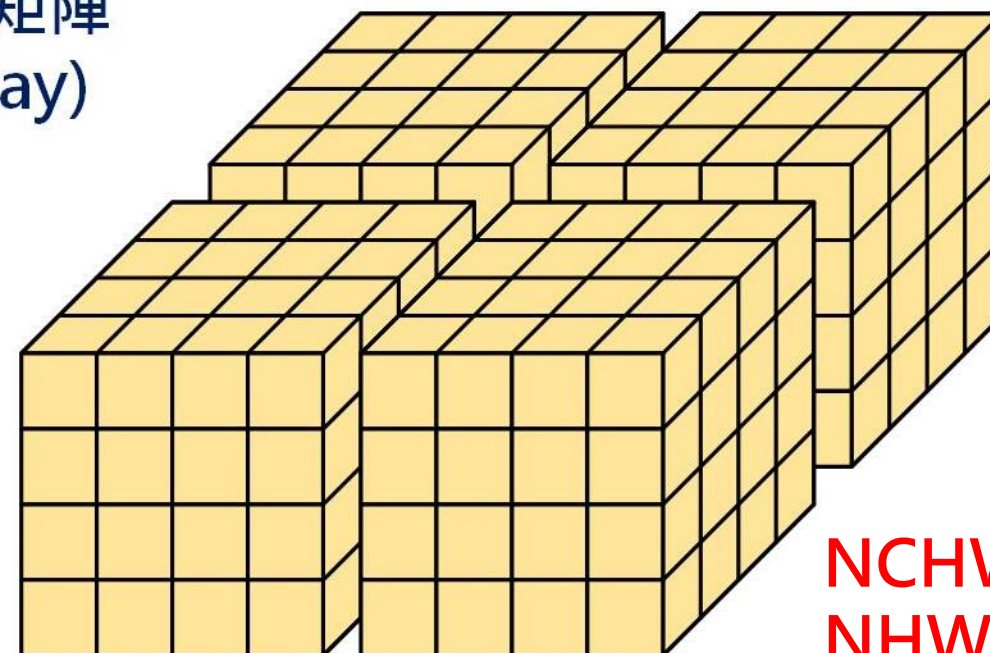


2維 矩陣
(array)

彩色影像
RGB ? BGR?



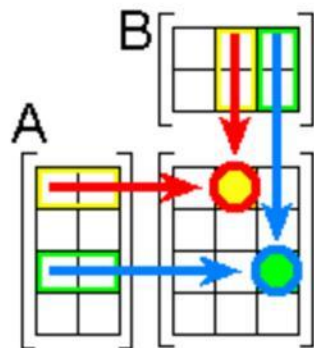
3維 張量(tensor)



4維 張量(tensor)

NCHW ?
NHWC ?

矩陣乘法



$$(AB)_{ij} = \sum_{r=1}^n a_{ir} b_{rj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj}$$

$$(AB)_{1,2} = \sum_{r=1}^2 a_{1,r} b_{r,2} = a_{1,1} b_{1,2} + a_{1,2} b_{2,2}$$

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ \vdots \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & \cdots \\ b_{2,1} & b_{2,2} & b_{2,3} & \cdots \\ b_{3,1} & b_{3,2} & b_{3,3} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} = [B_1 \quad B_2 \quad B_3 \quad \cdots]$$

$$\mathbf{AB} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ \vdots \end{bmatrix} \times [B_1 \quad B_2 \quad B_3 \quad \cdots] = \begin{bmatrix} (A_1 \cdot B_1) & (A_1 \cdot B_2) & (A_1 \cdot B_3) & \cdots \\ (A_2 \cdot B_1) & (A_2 \cdot B_2) & (A_2 \cdot B_3) & \cdots \\ (A_3 \cdot B_1) & (A_3 \cdot B_2) & (A_3 \cdot B_3) & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

平行 / 並行運算

平行(Parallelism)

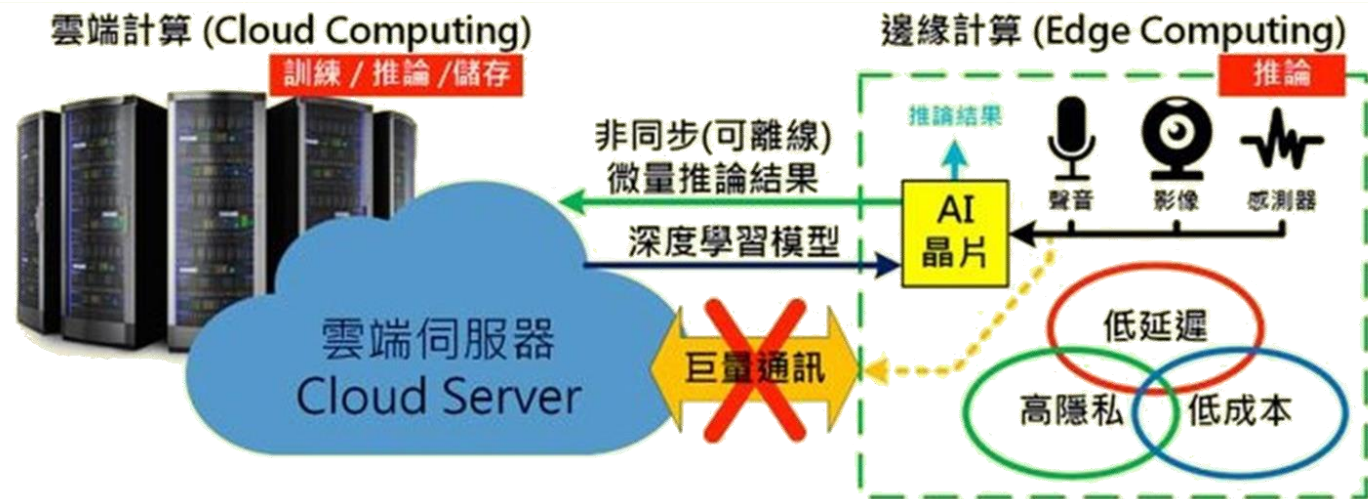
- 將工作分割成多份不同參數但內容相同工作內容，同時執行，待全部完成，再進行下一動作。
- 如矩陣乘法是由很多個 $A \times B + C$ 組合而成，傳統上須依序計算再組合結果，此時就很適合分割成很多小工作一起計算，再合併結果得到所需結果。

並行(Concurrency)

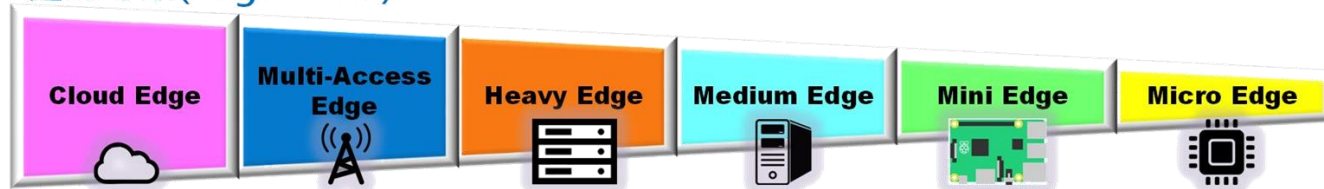
- 可將工作(task)拆分成多個獨立工作，內容可不相同，可同時運行，互不干擾，亦可不同時完成。
- 如編譯一個由很多函式組成的程式，利用多處理器（或多執行緒）方式同時編譯，待全部編譯完成後再連結成執行檔。



Edge AI

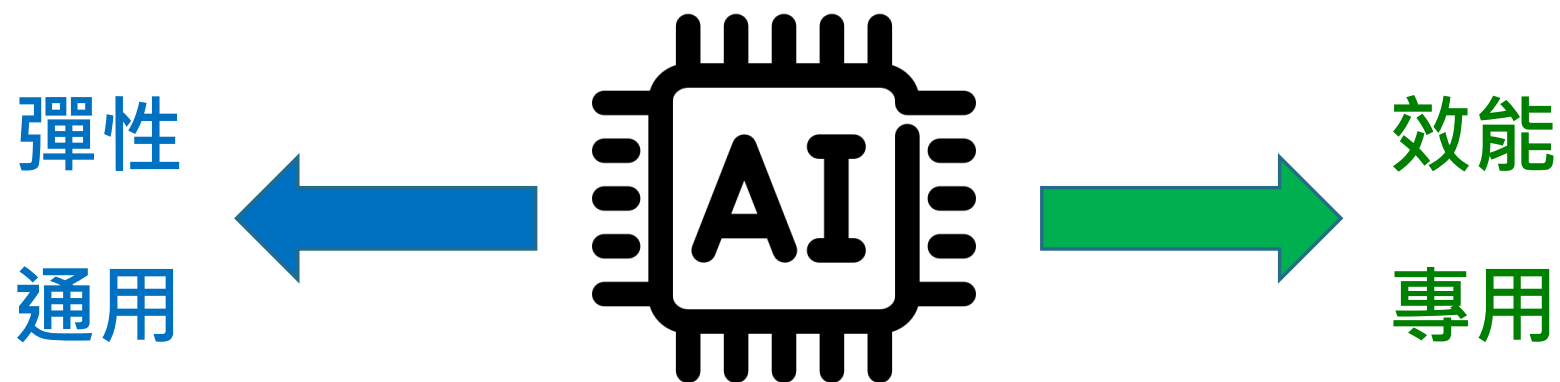


邊緣等級(Edge Level)



2.2. 加速運算晶片

加速運算晶片



CPU — 指令集架構(ISA)

複雜指令集 (CISC)

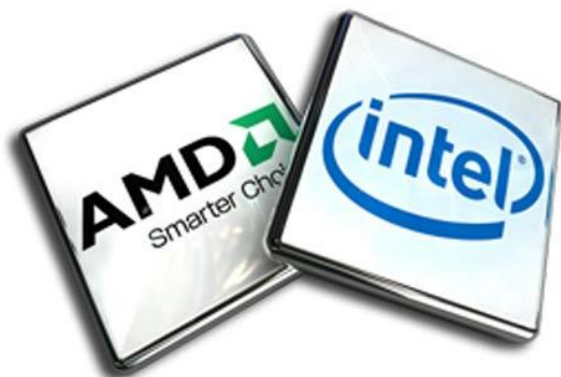
- INTEL(x86)
- AMD
- VIA

精簡指令集 (RISC)

- ARM
- RISC-V
- MIPS

自定義指令集

- RISC-V



arm



RISC-V

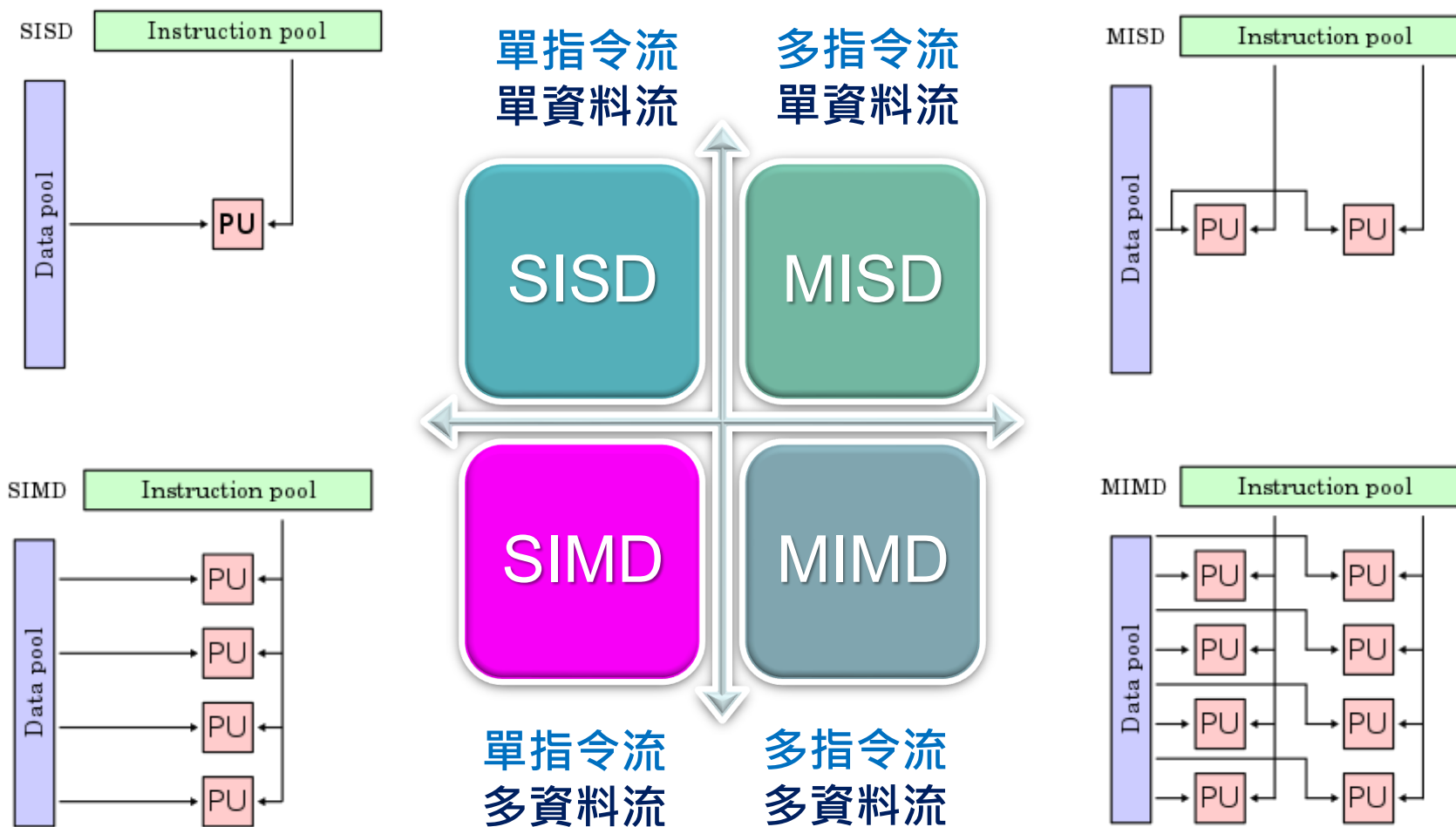
MIPS



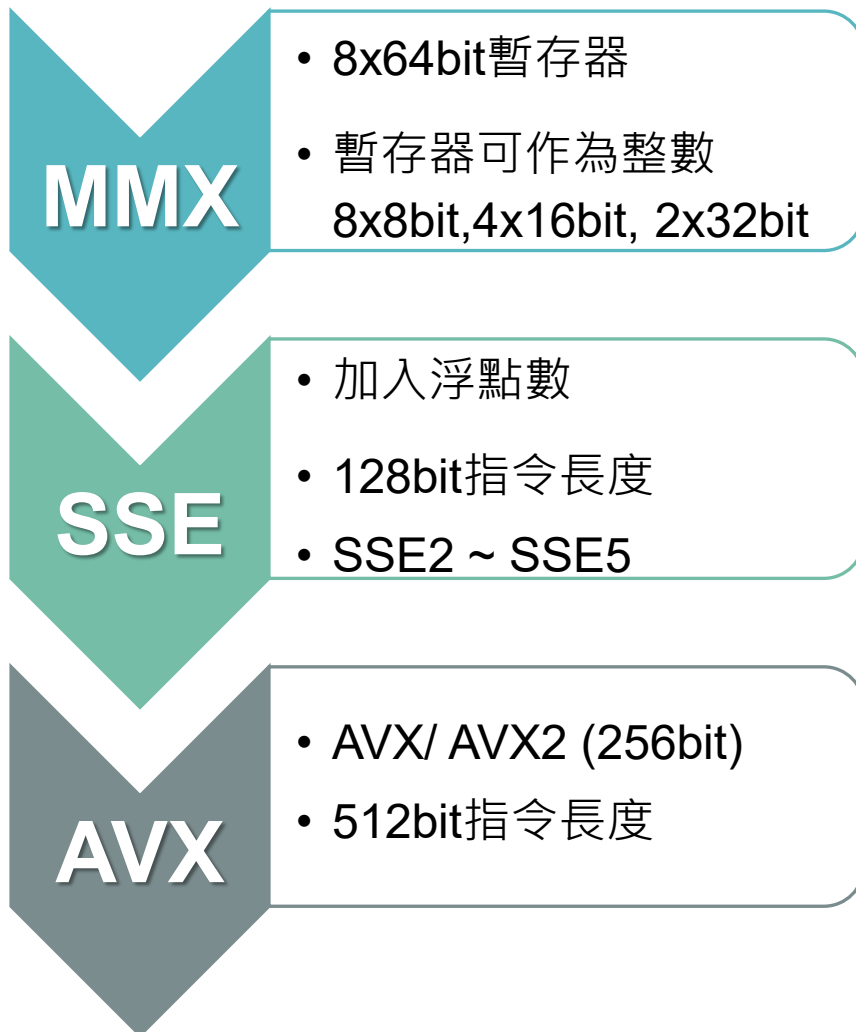
siFive

ANDES
TECHNOLOGY

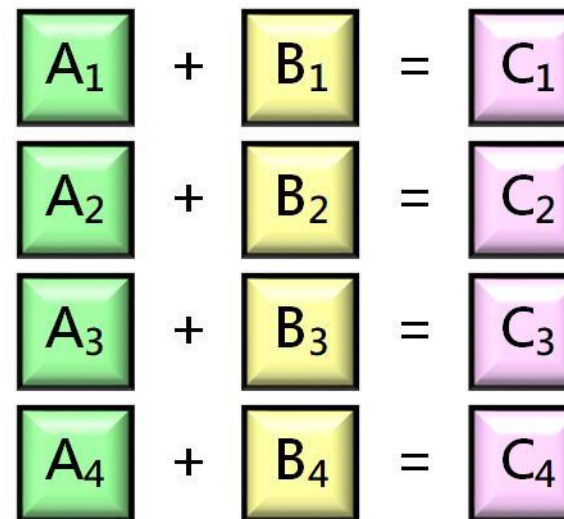
CPU—指令流與資料流



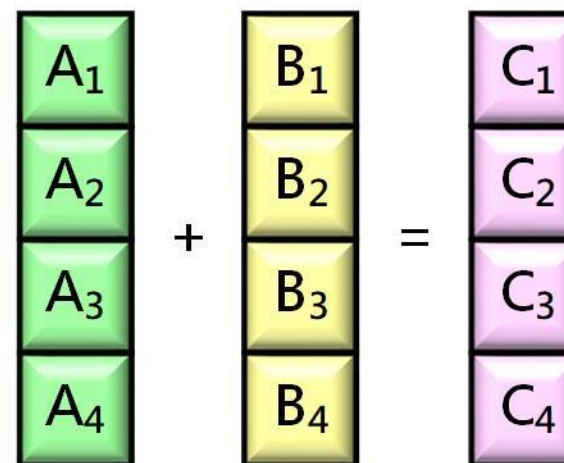
CPU — x86 SIMD



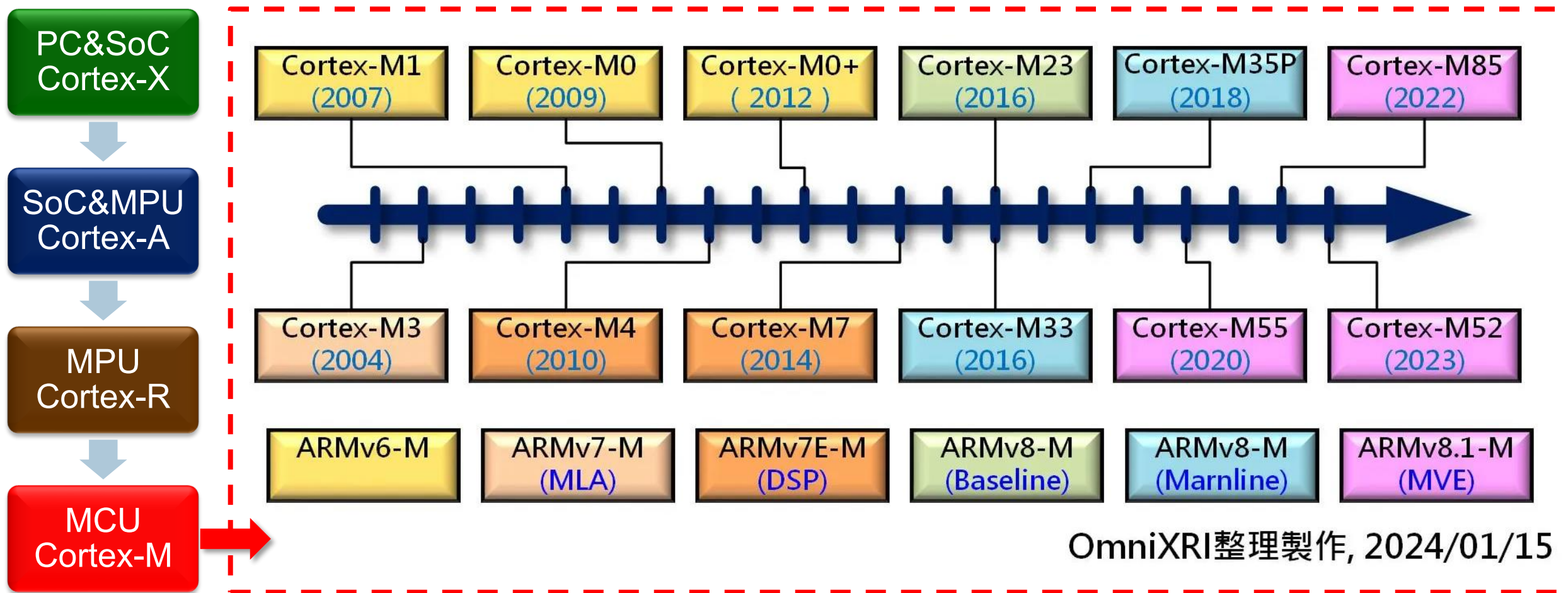
單指令流
單資料流
(SISD)



單指令流
多資料流
(SIMD)



CPU – Arm Cortex-M 加速指令集



OmniXRI整理製作, 2024/01/15

資料來源：<https://omnixri.blogspot.com/2024/01/vmaker-edge-ai-13-npuai.html>

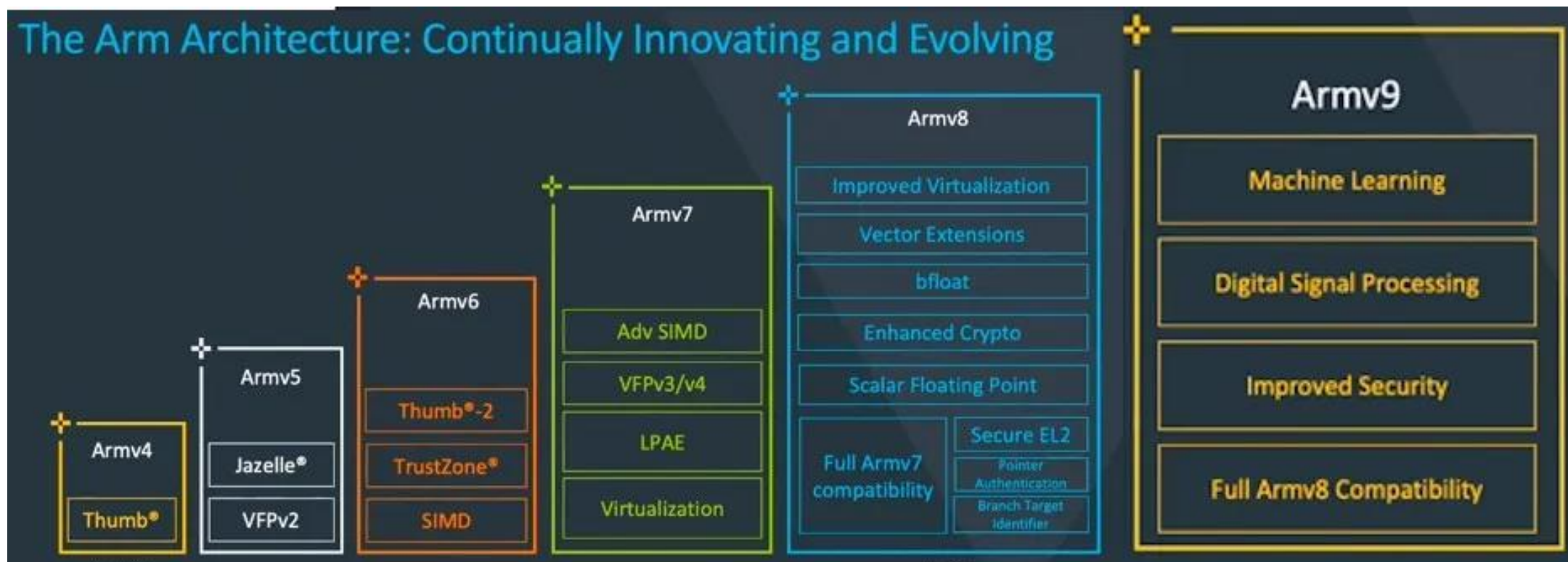
CPU — ARM Cortex-A 加速指令集

NEON Armv7/v8
Cortex-A/R SIMD

SVE Armv8-A
可伸縮向量擴展

SME Armv9-A
可伸縮矩陣擴展

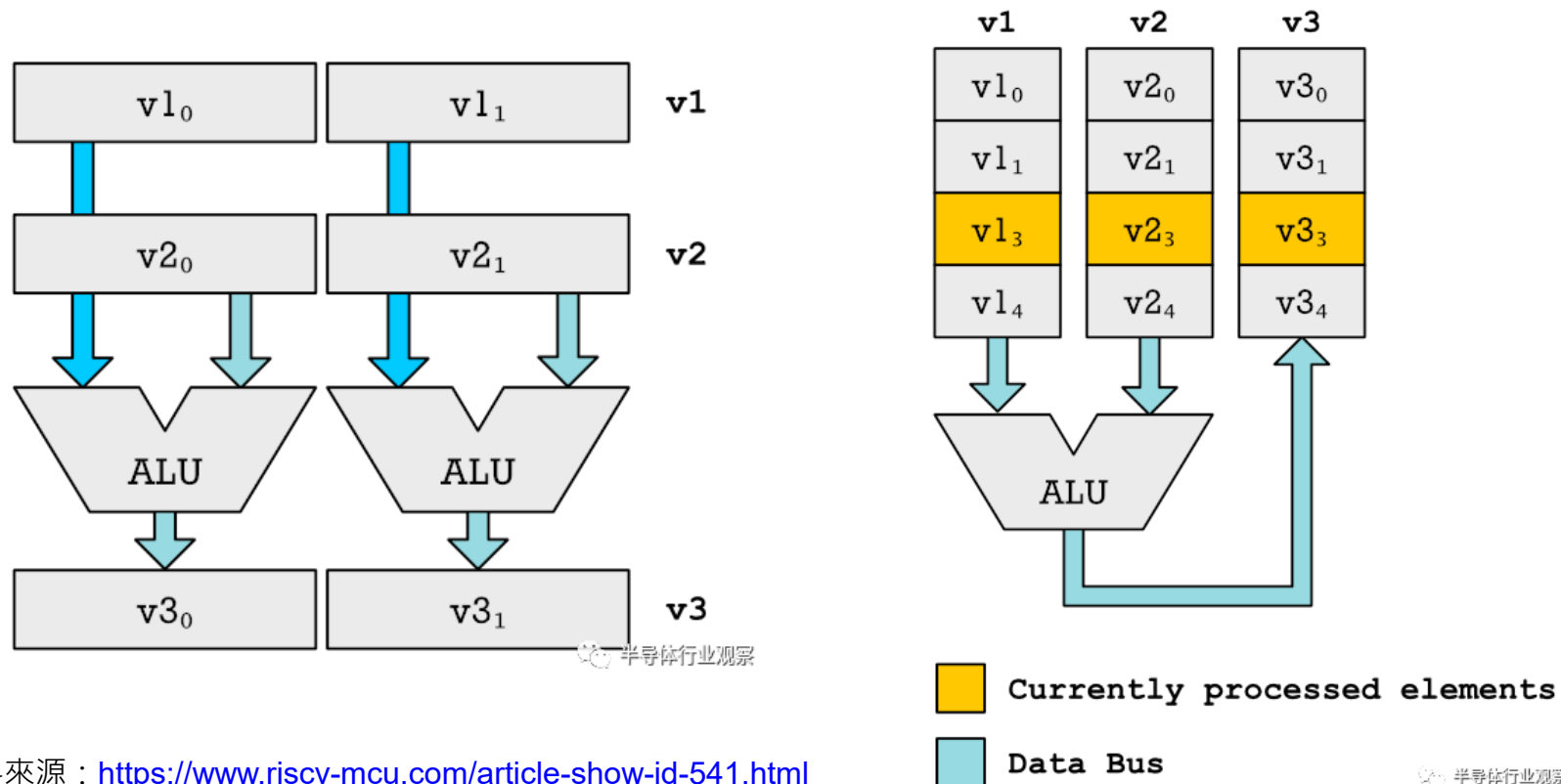
VPU Armv9-A
向量處理器



CPU – RISC-V 向量指令集

擴充指令集 V：向量運算

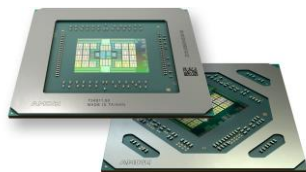
擴充指令集 P：單指令多資料流(SIMD)



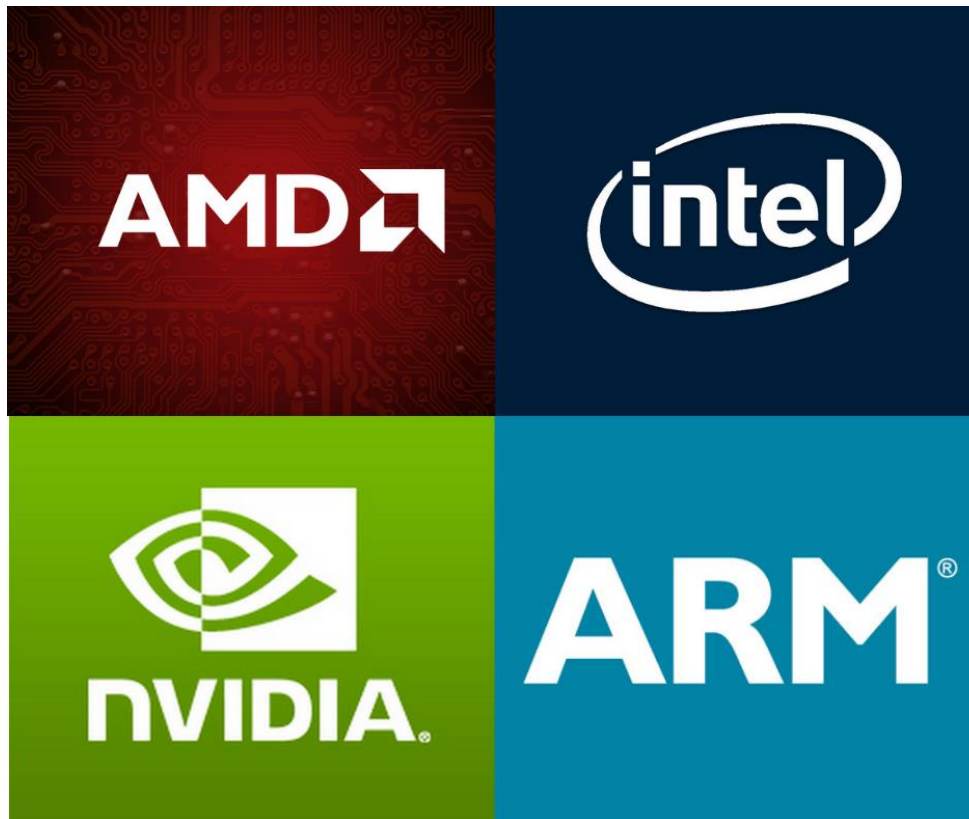
資料來源：<https://www.riscv-mcu.com/article-show-id-541.html>

GPU — 主要供應商

Radeon



GeForce
Quadro/Tesla
Jetson/DGX



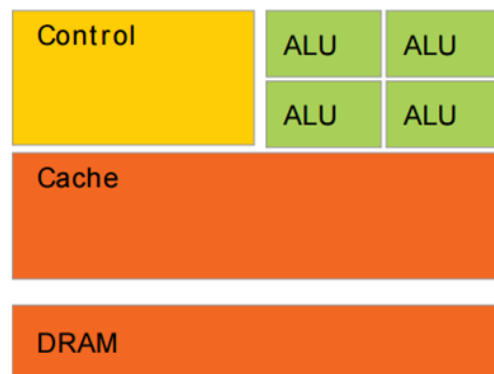
HD Graphic
Iris Xe / Arc



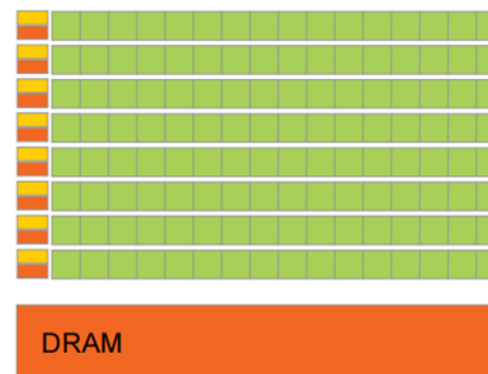
MALI



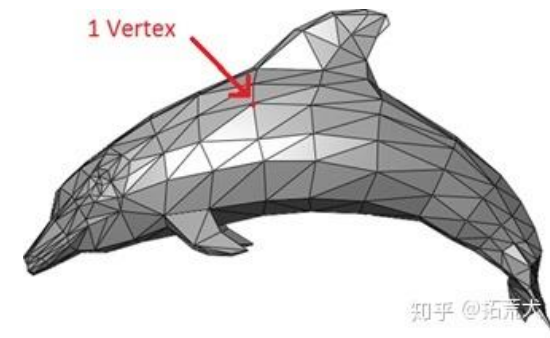
GPU — 結構與工作流程



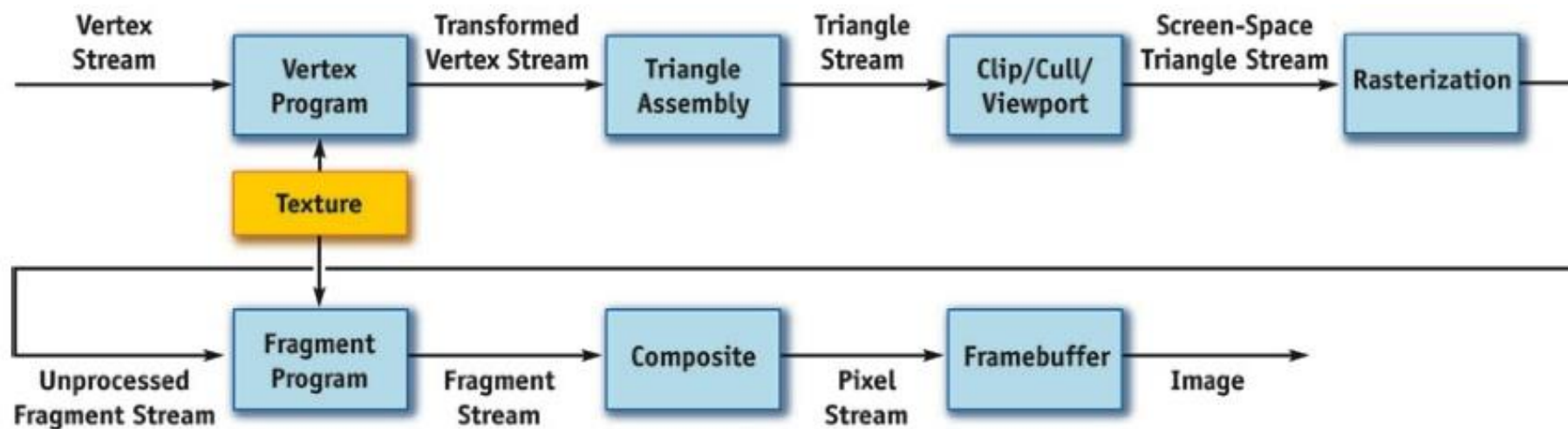
CPU



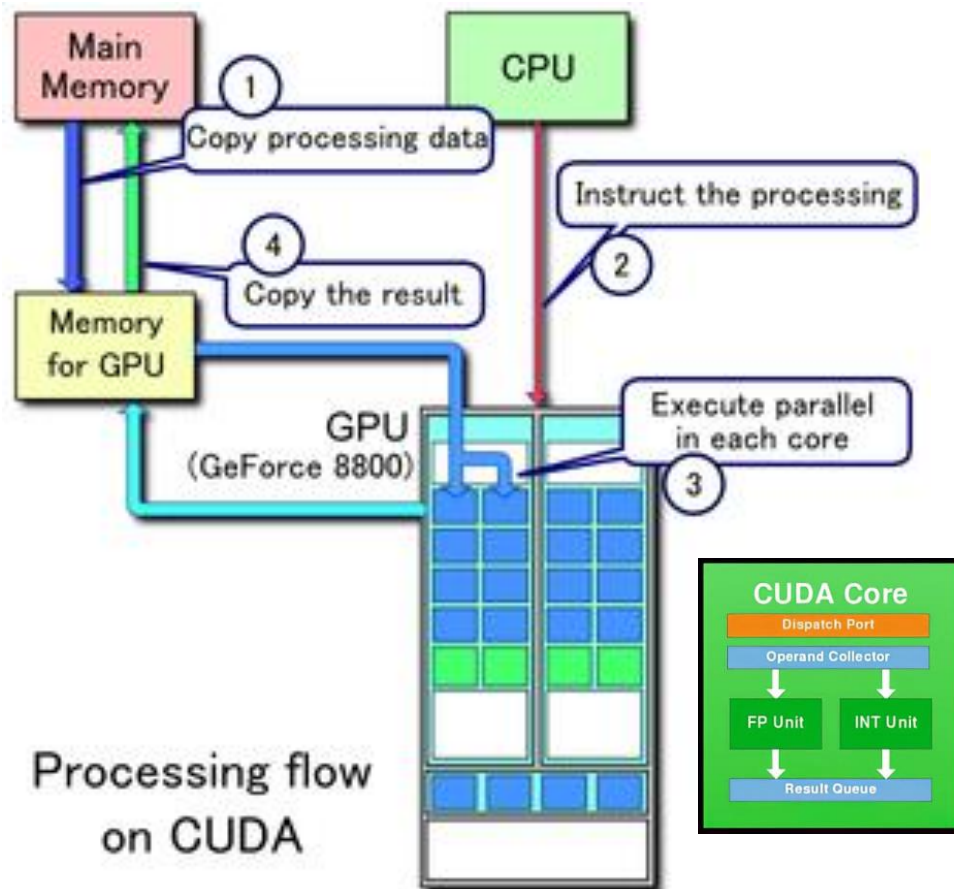
GPU



渲染(Render)

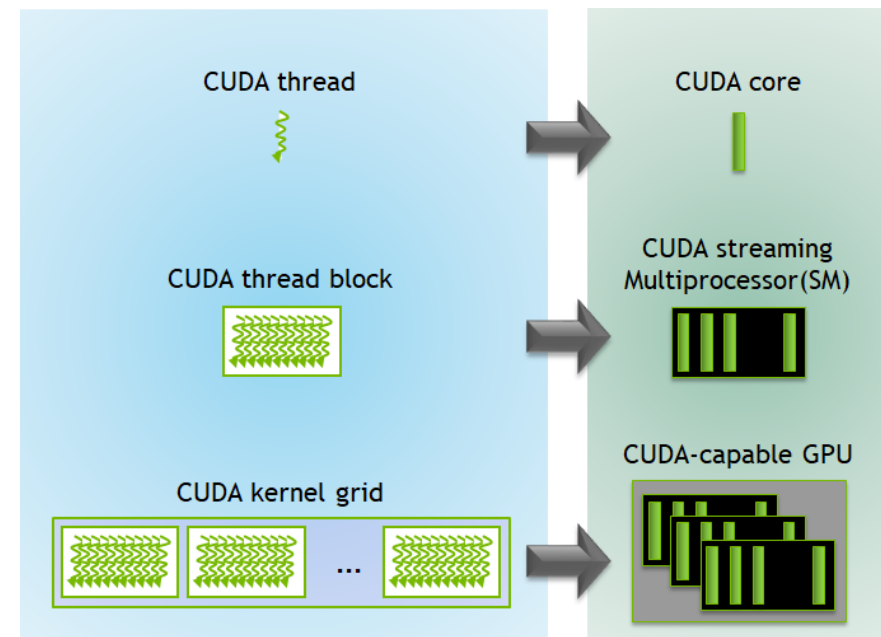


GPU – Nvidia Cuda Core



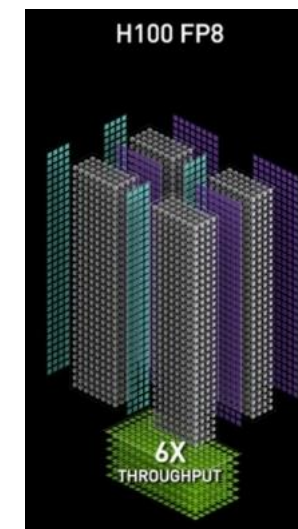
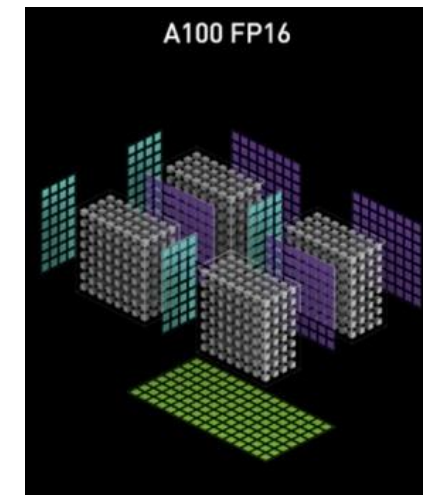
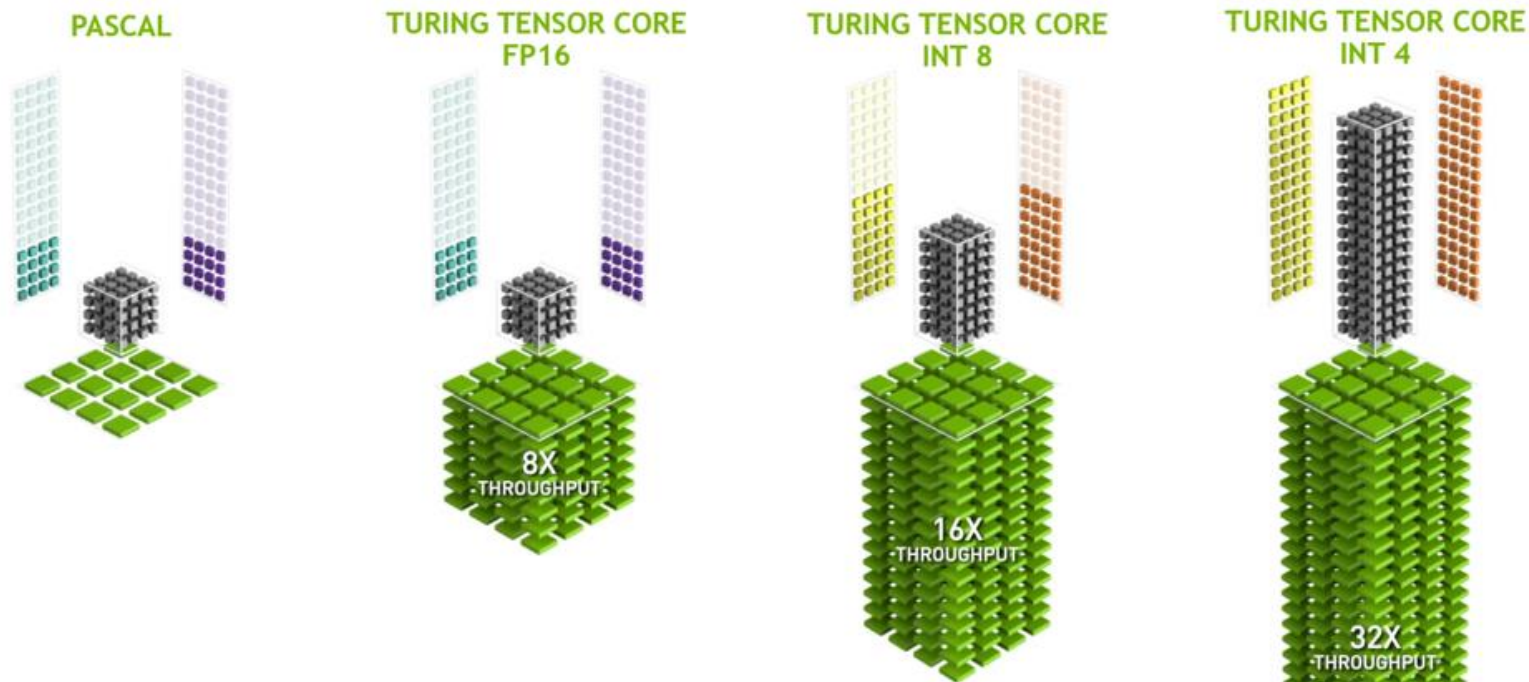
計算架構

➤ GPGPU / CUDA / OpenCL



資料來源：<https://zh.wikipedia.org/wiki/CUDA>

GPU — Nvidia Tensor Core



$$D = \begin{pmatrix} A_{0,0} & A_{0,1} & A_{0,2} & A_{0,3} \\ A_{1,0} & A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,0} & A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,0} & A_{3,1} & A_{3,2} & A_{3,3} \end{pmatrix} + \begin{pmatrix} B_{0,0} & B_{0,1} & B_{0,2} & B_{0,3} \\ B_{1,0} & B_{1,1} & B_{1,2} & B_{1,3} \\ B_{2,0} & B_{2,1} & B_{2,2} & B_{2,3} \\ B_{3,0} & B_{3,1} & B_{3,2} & B_{3,3} \end{pmatrix} = \begin{pmatrix} C_{0,0} & C_{0,1} & C_{0,2} & C_{0,3} \\ C_{1,0} & C_{1,1} & C_{1,2} & C_{1,3} \\ C_{2,0} & C_{2,1} & C_{2,2} & C_{2,3} \\ C_{3,0} & C_{3,1} & C_{3,2} & C_{3,3} \end{pmatrix}$$

FP16 或 FP32 FP16 FP16 或 FP32

GPU — Cuda / Tensor Core (1/2)

訓練用

NVIDIA GPU	A100	TU102	GV100	GP100
GPU架構	Ampere	Turing	Volta	Pascal
製程	TSMC 7nm	TSMC 12nm	TSMC 12nm	TSMC 16nm
晶片面積	826mm ²	754mm ²	815mm ²	610mm ²
電晶體數目	542億	186億	211億	153億
FP32 CUDA Cores	6912	4608	5120	3584
GPU Boost Clock	1.41GHz	1770MHz	1530MHz	1480MHz
記憶體時脈	2.4Gbps 5120bit	14Gbps 384bit	1.75Gbps 4096bit	1.4Gbps 4096bit
記憶體	40GB HBM2	24GB GDDR6	16/32GB HBM2	16GB HBM2
Single Precision (FP32)	19.5 TFLOPs	16.3 TFLOP	15.7 TFLOPs	10.6 TFLOPs
INT8 Tensor	624 TOPs	261 TOPs	N/A	N/A
FP16 Tensor	312 TFLOPs	130.5 TFLOPs	125 TFLOPs	N/A
TF32 Tensor	156 TFLOPs	N/A	N/A	N/A
TDP	400W	260W	300/350W	300W

GPU – Cuda / Tensor Core (2/2)

推論用

	NANO	TX2 NX	XAVIER NX*	ORIN NX 8 GB	ORIN NX 16 GB	AGX XAVIER**	AGX ORIN 32GB	AGX ORIN 64GB
GPU	128 Core Maxwell 0.5 TFLOPs (FP16)	256 Pascal cores 1.3 TFLOPS (FP16)	384 Core Volta 21 TOPs (INT8)	1024 Cores Ampere NVDLA V2 70 TOPS (INT8)	1024 Cores Ampere 2x NVDLA V2 100 TOPS (INT8)	512 Core Volta + NVDLA 10 TFLOPS (FP16) 32 TOPS (INT8)	1792 Cores Ampere 2x NVDLA V2 200 TOPS (INT8)	2048 Cores Ampere 2x NVDLA V2 275 TOPS (INT8)
CPU	4 core ARM A57	6 core Denver and A57 (2x) 2MB L2	6 core Carmel ARM V8 (3x) 2MB L2 + 4MB L3	6 cores A78 ARM V8 1.5MB L2 + 4MB L3	8 cores A78 ARM V8 2MB L2 + 4MB L3	8 core Carmel ARM V8 (4x) 2MB L2 + 4MB L3	8-core A78 ARM V8 2MB L2 + 4MB L3	12 core A78 ARM V8 3MB L2+ 6MB L3
Memory	4 GB 64-bit LPDDR4 25.6 GB/s	4 GB 128-bit LPDDR4 51.2 GB/s	8GB or 16GB 128-bit LPDDR4x 51.2 GB/s	8 GB 128-bit LPDDR5 102.4 GB/s	16 GB 128-bit LPDDR5 102.4 GB/s	32GB or 64 GB 256-bit LPDDR4x 137 GB/s	32 GB 256-bit LPDDR5 204.8 GB/s	64 GB 256-bit LPDDR5 204.8 GB/s
Storage	16 GB eMMC	16 GB eMMC	16 GB eMMC	- (Supports external NVMe)	- (Supports external NVMe)	32 GB eMMC	64 GB eMMC	64 GB eMMC
Encode	4K @ 30 (H.265)	4K @ 60 (H.265)	2x 4K @ 30 (H.265)	1x 4K @ 60 (H.265)	1x 4K @ 60 (H.265)	4x 4K @ 60 (H.265)	1x 4K @ 60 (H.265)	2x 4K @ 60 (H.265)
Decode	4K @ 60 (H.265)	2x 4K @ 60 (H.265)	2x 4K @ 60 (H.265)	2x 4K @ 60 (H.265)	2x 4K @ 60 (H.265)	6x 4K @ 60 (H.265)	2x 4K @ 60 (H.265)	3x 4K @ 60 (H.265)
Camera	12 (3x4 or 4x2) MIPI CSI-2 D-PHY 1.1 lanes (18 Gbps)	12 lanes (3x4 or 5x2) MIPI CSI-2 D-PHY 1.2 (30 Gbps)	12 lanes (3x4 or 6x2) MIPI CSI-2 D-PHY 1.2 (30 Gbps)	8 lanes MIPI CSI- 2 D-PHY 1.2 (20 Gbps)	8 lanes MIPI CSI- 2 D-PHY 1.2 (20 Gbps)	16 lanes MIPI CSI-2 8 lanes SLVS-EC D-PHY (40 Gbps) C-PHY (59 Gbps)	16 lanes MIPI CSI-2 D-PHY 2.1 (40 Gbps) C-PHY 2.0 (164 Gbps)	16 lanes MIPI CSI-2 D-PHY 2.1 (40 Gbps) C-PHY 2.0 (164 Gbps)
Mechanical	69.6mm x 45mm 260 pin edge connector	69.6mm x 45mm 260 pin connector	69.6mm x 45mm 260 pin edge connector	69.6mm x 45mm 260 pin edge connector	69.6mm x 45mm 260 pin edge connector	100mm x 87mm 699 pin connector	100mm x87 mm 699 pins connector	100mm x87 mm 699 pins connector
Software	JetPack SDK - Unified software release across all Jetson products							

資料來源：<https://www.seeedstudio.com/blog/nvidia-jetson-comparison-nano-tx2-nx-xavier-nx-agx-orin/>

DSP — 常見供應商

數位信號處理器(Digital Signal Processor)



cā d e n c e

SYNOPSYS®



TI

OMAP
C6000/5000/
2000
DaVinci

Qualcomm

Hexagon

Cadence

Tensilica
* Vision P5/
P6 / C5
* Xtensa
Lx6 / Lx7

Synopsys

Arc
* EMxD
* HS4xD
* VPX

Ceva

* XC
* BX
* XM6
* NeuPro-M

(**ESP32**)

DSP — 主要功能

常見聲音及影像應用運算

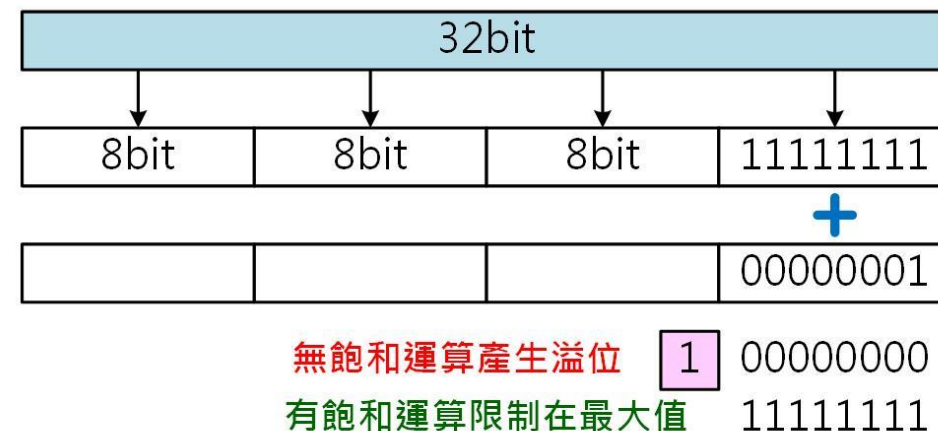
- FIR、IIR、DFT、DCT、FFT、Dot product、Convolution及其它矩陣運算等。
- 可進行整數、定點數、浮點數計算。

乘積累加 (Multiply Accumulate, MAC)

- 乘法和加法一次作完，如 $y = ax + b$

飽和運算 (Saturation Arithmetic)

- 執行SIMD指令，將多位元指令分拆為少位元運算時，避免溢位產生。



NPU — 主要供應商

- 以神經網路運算為主要工作的晶片，各廠家稱呼方式略有不同，如 TPU, APU, AI Accelerator 等。
- 可支援通用型開發框架及模型，如 ONNX, TensorFlow, PyTorch 等。



NPU – MCU級常見供應商

➤ arm Ethos-U55

ALIF Ensemble, Nuvoton M55,
Himax WE2, Infineon PSoC Edge,
Ambiq, Synaptics Astra SR

➤ arm Ethos-U65

NXP i.MX93

➤ arm Ethos-U85

ALIF E8

註：常見算力單位 **TOPS** (每秒一兆次，
 10^{12})，要注意數值格式為 **FP32 / FP16 /**
INT16 / INT8 / FP8 / FP4。

➤ ST Neural-ART Accelerator

STM32N6

➤ NXP eIQ Neutron NPU

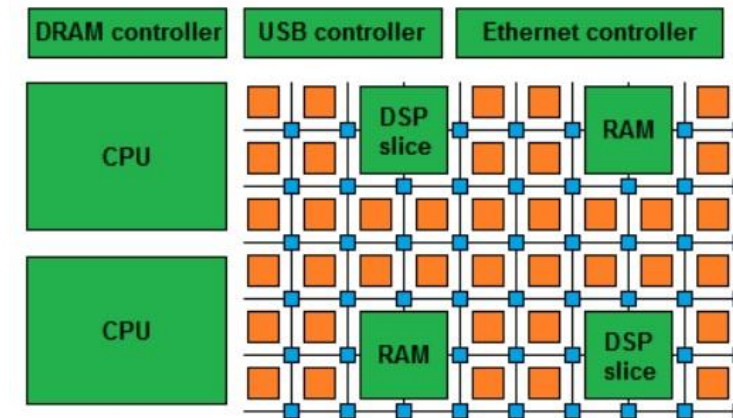
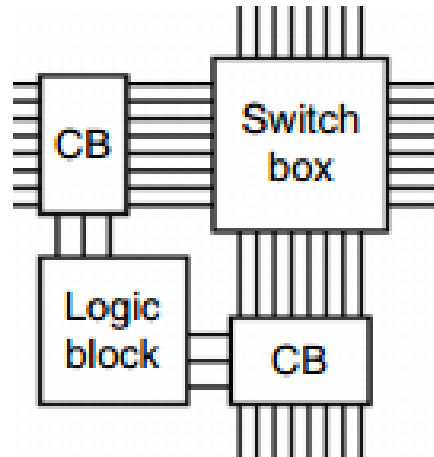
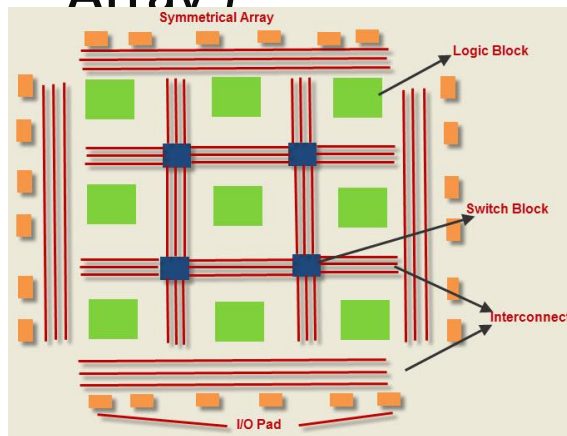
i.MX95, MCX-N54/94

➤ Renesas DRP-AI

RZ/V2M, RZ/V2L, RZ/V2H, RZ/V2N

FPGA

現場可程式化邏輯陣列(Field - Programmable Gate Array)



AMD (Xilinx)



Altera(原Intel)



Microsemi

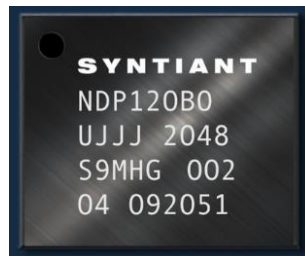


Lattice



Microchip

其它類型 — 記憶體內計算 (CiM)



SYNTIANT
NDP120

直接在記憶體單元內完成邏輯運算或數學計算，數據不需要離開記憶體，解決了搬運數據的功耗與速度問題。



Witmem 知存科技
WTM2101

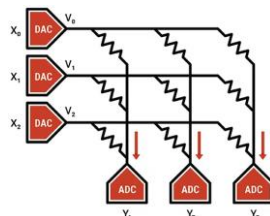
Compute in Memory 別稱

- * Processing-in-Memory (PiM)
- * Computational Memory
- * In-Memory Accelerator
- * Near-Memory Computing (NMC)
- * 存算一體（大陸常用稱呼）

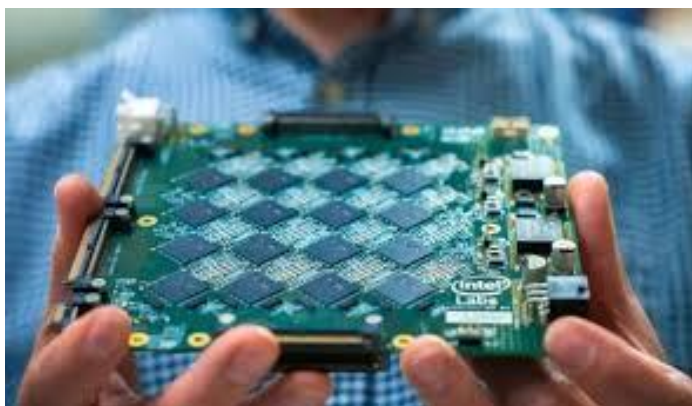
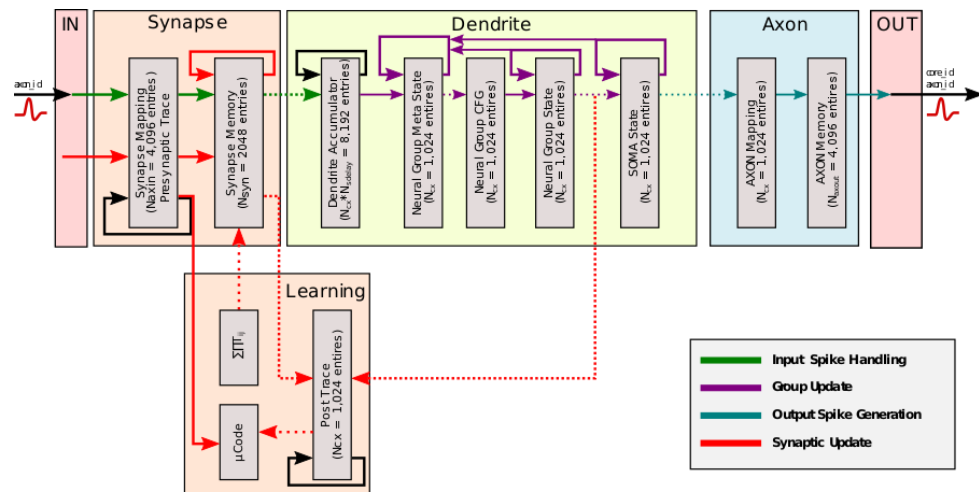


MYTHIC
M1076 Analog
Matrix Processor

常見實現技術：SRAM, DRAM, Flash, RRAM, MRAM等。



其它類型 — 類腦晶片



INTEL Pohoiki Beach
(64 x Loihi Chip)

Neuromorphic 神經形態

➤ 數位式

- IBM TrueNorth
- INTEL Loihi
- SpiNNaker

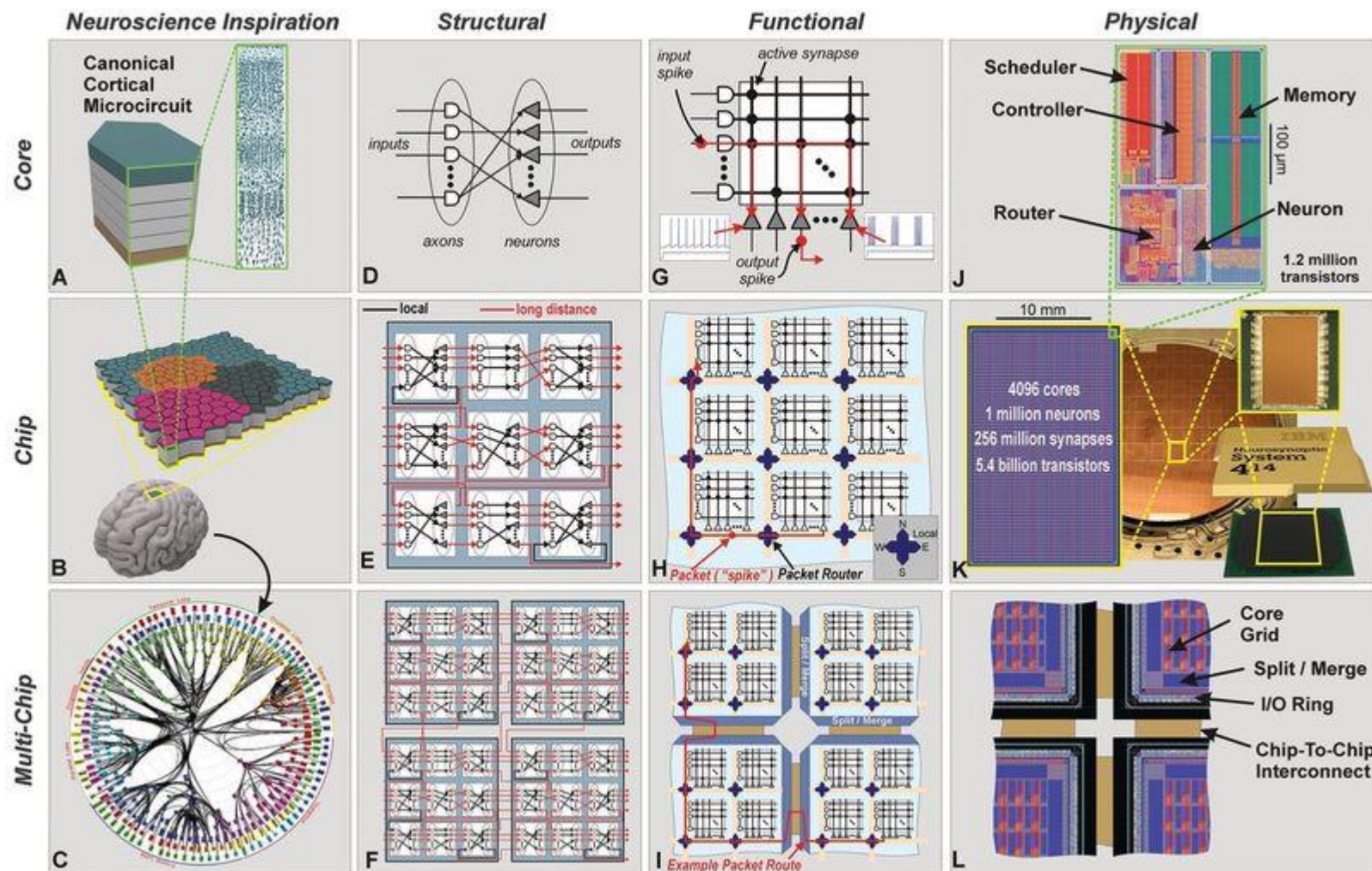
➤ 類比式

- Neurogrid
- BrainScales
- ROLLS

➤ 新材料

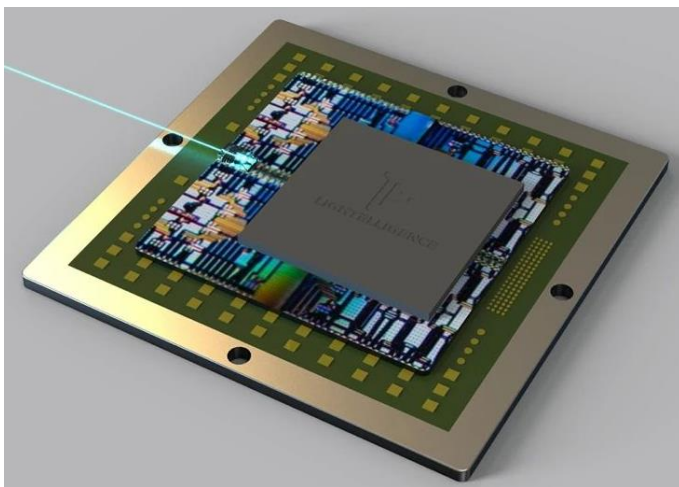
- Memristor

其它類型 — 類腦晶片 IBM TrueNorth



資料來源：<https://www.greencarcongress.com/2014/08/20140808-truenorth.html>

其它類型—光子晶片



Lightmatter

- MIT獨立出新創公司
- 利用矽光子和微機電製程技術，以毫瓦等級雷射光供電，能以接近光速執行矩陣向量乘法。

曦智 lightelligence

- 大陸團隊，百度投資
- 和Lightmatter皆為MIT技術，技術類似。

資料來源：<https://www.eettaiwan.com/20201006nt31-optical-compute-promises-game-changing-ai-performance/>

資料來源：<https://www.lightelligence.co/>

硬體加速方式與限制

矩陣計算核心 $Y = AX + B$



1. CPU 加速工作時脈
2. 單指令週期硬體乘法、除法、浮點數計算
3. 使用 DSP 指令 (乘法累加MAC)
4. 平行 / 向量指令集 (SIMD, MVE, NEON)
5. 多核心加速 (同質、大小核)
6. GPU 加速器 (CUDA, Tensor, NVDLA)
7. DSP / NPU / FPGA 加速器
8. 專用晶片 ASIC

硬體加速阻礙

算力牆

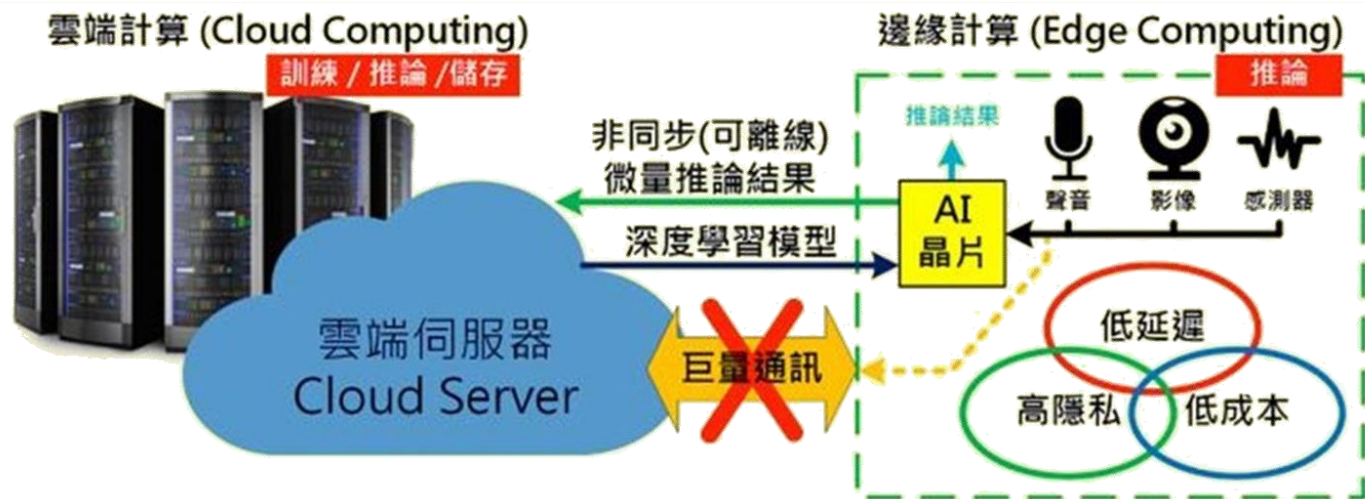
記憶體牆

能耗牆

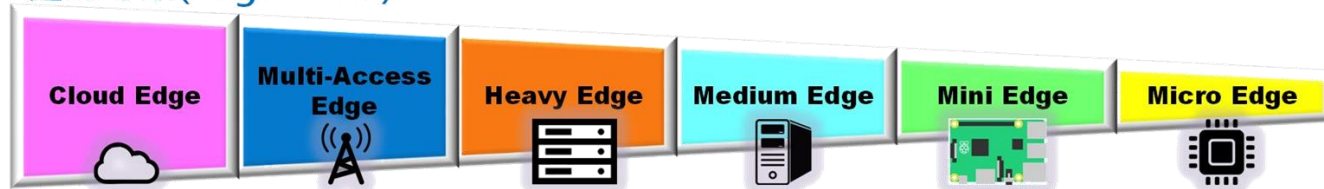
通訊牆



Edge AI



邊緣等級(Edge Level)



2.3. 嵌入式級硬體

各式開發板外型



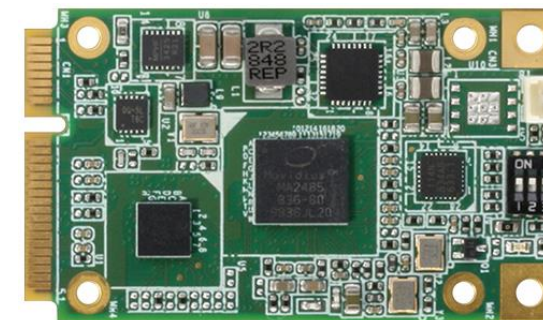
Pi Style



USB Dongle



Pin



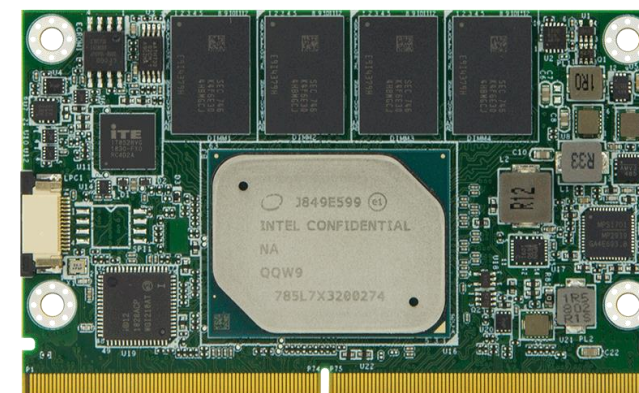
mPCIe / M.2



Mini ATX



SOM



SMARC

迷你電腦 — INTEL 系列



AAEON UP Squared
(Pentium N4200 / Celeron N3350)



AAEON UP Extrme
(Core i7/i5/i3/Celeron)



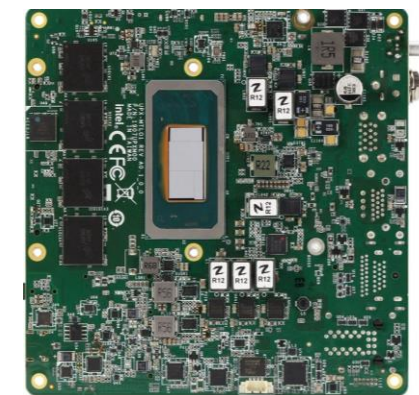
AAEON COM-TGUC6
(Tiger Lake UP3 with GPU Iris Xe)



Intel Arria 10 SoC Board
(FPGA Arria 10)



AAEON BOXER-8320AI
(Core i3-6100U + Dual Myriad VPUs)



AAEON UP Xtreme ARL
Arrow Lake Core Ultra 200H

單板微電腦 – NVIDIA

Jetson Orin 電腦視覺、生成式AI應用



Jetson Thor 機器人應用



Jetson DGX Spark 生成式AI訓練、推論

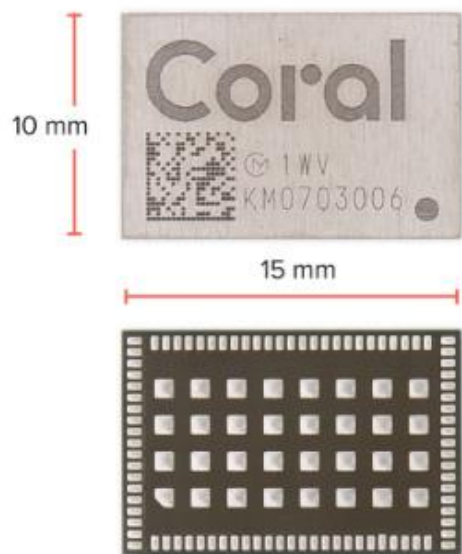


項目	Orin Nano	Orin NX	AGX Orin	AGX Thor	DGX Spark (GB10)
CPU	Arm Cortex-A78AE x6	Arm Cortex-A78AE x8	Arm Cortex-A78AE x12	Arm Neoverse-V3AE x14	Arm 10P + 10E
LPDDR5	8GB	16GB	64GB	128GB	128GB
CUDA core	1024(Ampere)	1024(Ampere)	2048(Ampere)	2560(Blackwell)	6144(Blackwell)
算力	80TOPS(INT8)	100TOPS(INT8)	275TOPS(INT8)	2070TFLOPS(FP4)	1 PetaFLOP (FP4)

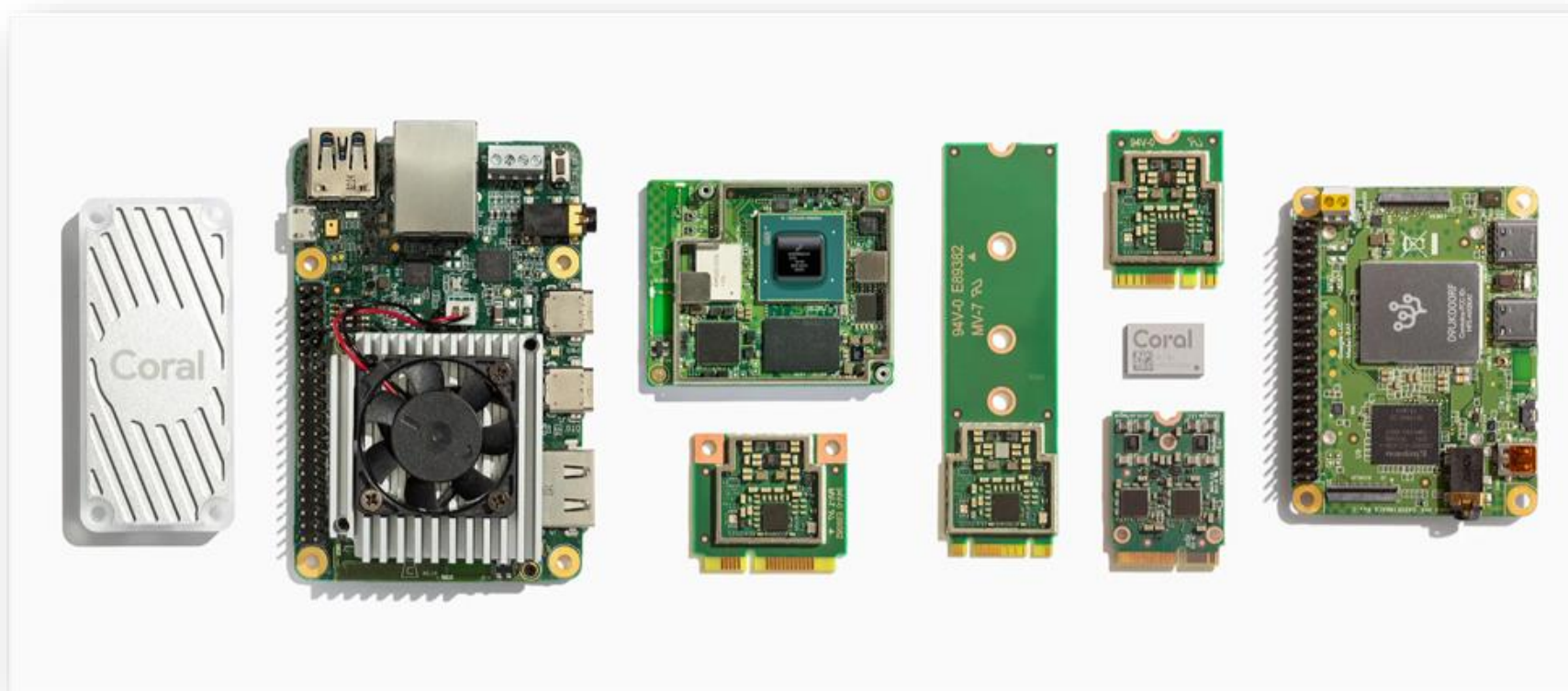
單板微電腦 — GOOGLE

Google Coral (Edge TPU) Series

ASUS IoT

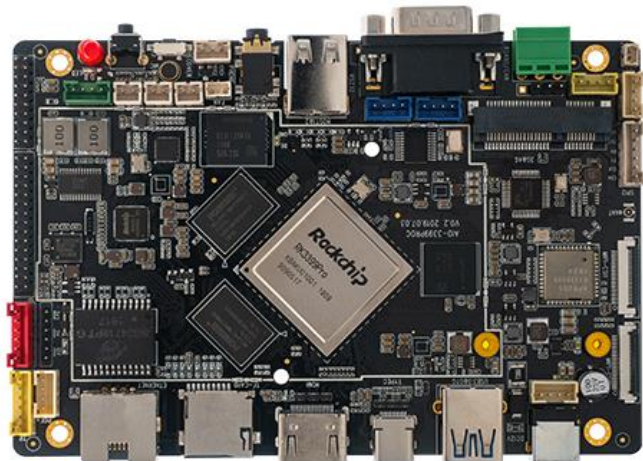


Accelerator Module
4 TOPs (int8)
2 TOPs pre Watt



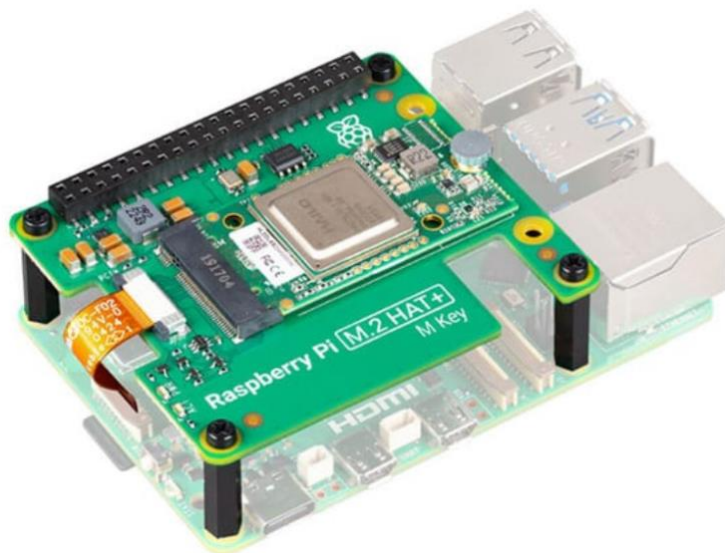
資料來源：<https://iot.asus.com/tw/gpu-edge-ai-accelerators/google-tpu/>

單板微電腦 – Arm Based



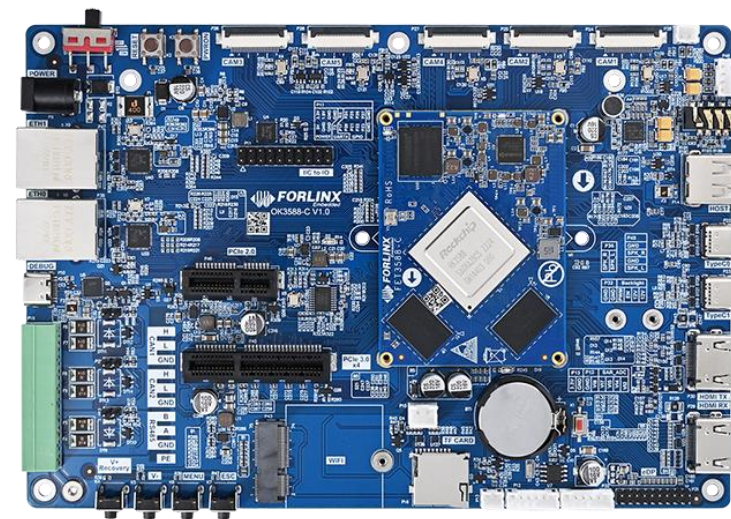
Rockchip RK3399Pro

Cortex-A72 *2 + Cortex-A53 *4
NPU 3 TOPS(INT8/16)
GPU Mali-T860MP4



Raspberry Pi 5 + AI Kit

Boardcom BCM2712
Arm Cortex-A76 x4
AI Kit – Hailo-8L13 TOPS(INT8)
AI Hat+ – Hailo-8 26 TOPS(INT8)
AI HAT+2 – Hailo-10H 40TOPS(INT4)



SBC3800

Rockchip RK3588
Arm Cortex-A76 x4 + A55 x4
Arm GPU Mali-G610 MC4
NPU 6 TOPS

單板微電腦 – 台灣 AI 開發板

凌陽 (Sunplus) – SP7350

Sunplus C3V SP7352 Demo Kit



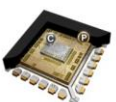
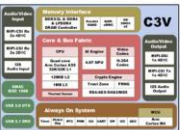
Banana Pi BPI-F4 (SP7350)



<https://w3.sunplus.com/tw/products/plus1.asp>

C3V

- Arm Cortex-A55 4核(@2.1GHz)
- Arm Cortex-M4@400MHz
- 284KB SRAM
- AI Engine(NPU) 4.6TOPS@1GHz
- DDR4 up to 8GB
- H.264 Codec

耐能 (Kneron) – KLM730A

KL520 2280 card KL720 EVB




KL720 AI加速棒



KNEO-Pi



KNEO-Pi



<https://kneron.github.io/kneopi-documentation/index.html>

KNEO-Pi

- Arm Cortex-A55 4核
- KDP730 NPU (4eTOPS@INT8)
- DSP (AI co-processor)
- MJPEG/H.264/H.265 Decoder
- 樹莓派相容尺寸及GPIO排針
- 可支援2組CSI及1組DSI

聯發科 (MediaTek) – Genio IoT

AAEON SMARC uCOM-M700/M510



AMobile SOM SM1200



Genio 系列

MEDIA TEK

晶片型號	主要核心	NPU	主要應用
Genio 1200	4x CA78 + 4x CA55 (2.2GHz)	4.8 TOPs	旗艦 AIoT、工業 HMI
Genio 720	2x CA78 + 6x CA55 (2.2GHz)	10 TOPs	生成式 AI、智慧零售
Genio 700	2x CA78 + 6x CA55 (2.2GHz)	4 TOPs	工業自動化、4K 數位看板
Genio 520	2x CA78 + 6x CA55 (2.0GHz)	10 TOPs	主流 AIoT、邊緣運算
Genio 510	2x CA78 + 4x CA55 (2.0GHz)	3.2 TOPs	進階 IoT、中階多媒體
Genio 350	4x CA53 (2.0GHz)	0.35 TOPs	入門智慧家庭、POS、Kiosk

Mediatek Genio 510 EVK



VIA VAB-3000 Genio 350 EVK



<https://www.mediatek.com/zh-tw/products/iot/genio-iot>

註：CA=Arm Cortex-A

聯詠 (Novatek) – Smart Vision SoC

Novatek NT98530 Novatek NT98692




Novatek NT98560 Novatek NT98566




NT985xx/6xx 系列

NOVATEK

主要用於智慧監控及數位影像錄製

晶片型號	主要核心	AI算力	主要用途
NT98692	4x CA73	6.4 TOPS	8K 邊緣運算攝影機、後端 XVR
NT98530	4x CA53	2.0 TOPS	高階 4K 安防監控 (IPC)
NT98529	2x CA9	1.5 TOPS	中高階 IPC、人臉辨識
NT98568	2x CA7	0.5 TOPS	電池型門鈴、智慧家居
NT98566	1x CA9	0.7 TOPS	Wi-Fi 6 智慧相機
NT98632	2x CA53	0.75 TOPS	NVR 錄影機、網路監控
NT98560	1x CA9	0.7 TOPS	主流級影像監控

註：CA=Arm Cortex-A

https://www.novatek.com.tw/zh-TW/Product/product/Index/system_single_chip_solution/SoC

晶心 (Andes) – QL7200 (RISC-V)

AXELERA Metis M.2 AI Accelerator (AndesCore AX25)



Novatek NT98692 (AndesCore NX27V)



ASUS Tinker V (Renesa RZ/Five - AndesCore AX45MP)



Andes QL7200 QiLai / Voyager (AX45MP+NX27V)



QiLai + Voyager

ANDES

QiLai(奇萊) 系統晶片

- RISC-V AX45MP 4核(@2.2GHz)
- NX27V 向量處理器(@1.5GHz)
- 512bit VLEN, DLEN

Voyager 開發板

- QiLai SoC
- 16GB DDR4, 16Mb SPI Flash
- 支援 AndeSight, AndeSoft, AndesAIRE NN SDK

<https://www.andestech.com/tw/產品與解決方案/andesaire-ai/>

華碩(Asus) – Tinker Board AI 開發板

Tinker Edge 3S (RK3566 0.8TOPS)



Tinker Edge 3N Plus (RK3568 0.8TOPS)



Tinker Board

ASUS

tinkerboard

類樹莓派單板微電腦

包含各種 Arm / RISC-V 微處理器

部份具有 NPU 或 Google EdgeTPU

可加速 AI 推論計算

Tinker Edge V (Renesa RZ/Five (AX45MP))



Tinker Edge R (RK3399PRO 3TOPS)



Tinker Edge T (iMX 8M + EdgeTPU 4TOPS)



Tinker Edge T (iMX 8M 4TOPS)



<https://tinker-board.asus.com/tw/index.html>

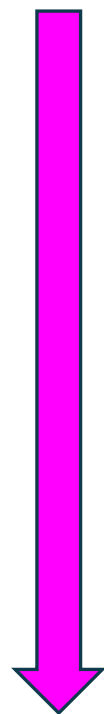
常見單晶片開發板

主要分類：

- MCU (SIMD, MVE指令集)
- MCU + NPU
- MCU + DSP
- MPU + GPU
- FPGA

算力

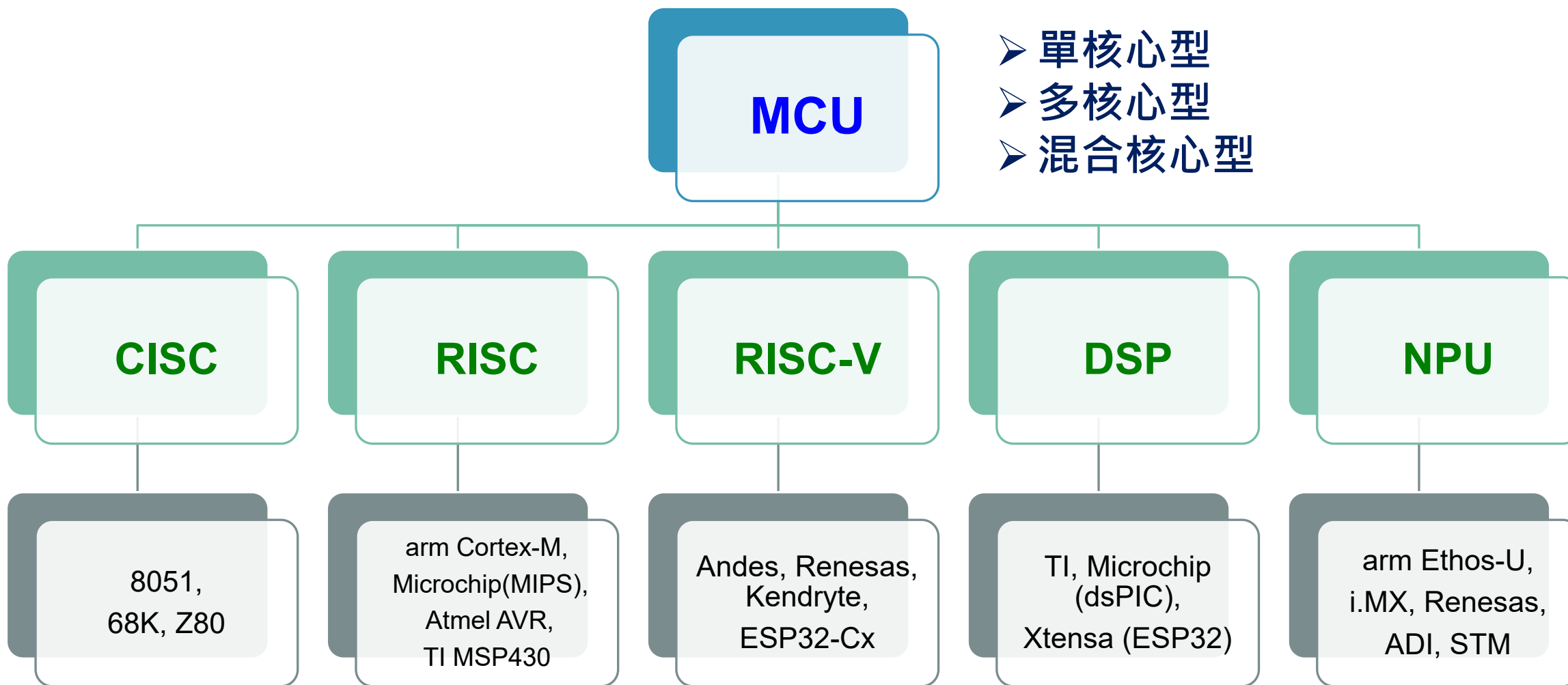
低



高



MCU 等級主要核心晶片分類



智慧感測型開發板(MCU SIMD 指令集)



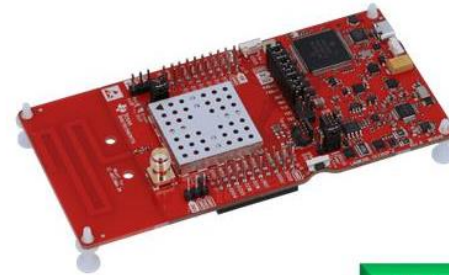
M0+ x2

Raspberry
Pi Pico (RP2040)



M4

Arduino
Nano 33 IoT /
BLE / BLE Sense



M4

TI
CC1352P Launchpad



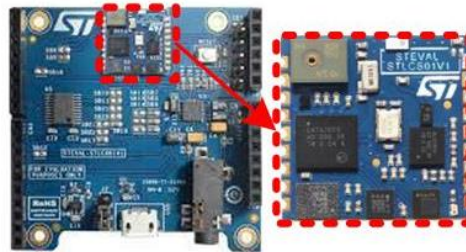
M4

Seeed
Xiao BLE / BLE Sense



M33

Silicon Labs
xG24-DK2601B



M4

ST
SensorTile
(STEVAL-STLKT01V1)



M4

Coretronic MEMS
CoreMaker-01



Syntiant
Tiny ML Board

資料來源：<https://omnixri.blogspot.com/2022/07/20220731coscupaimcutinyml.html>

OmniXRI整理製作, 2022/7/31

MCU / MCU+NPU 開發板

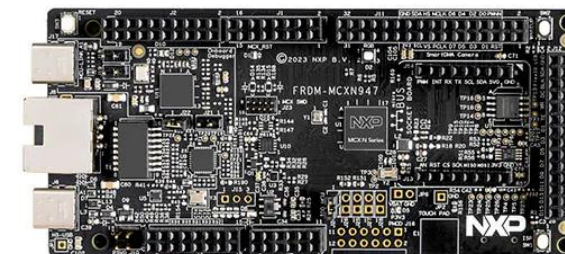
Seeed Wio Terminal

- Cortex-M4F x1



NXP MCX-N FRDM-MCXN947

- Cortex-M33 x2
- eIQ NPU x1



Nordic nRF9160 SiP Thingy:91

- Cortex-M33 x1



Raspberry Pi Pico 2W

- Cortex-M33 x2
- RISC-V Hazard3 x2



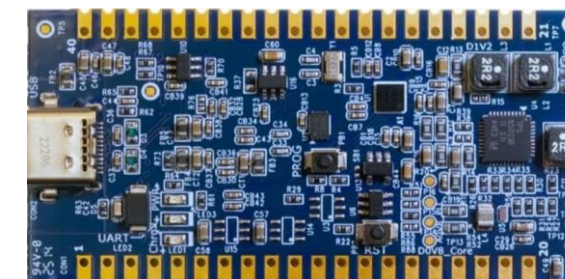
Arduino Nicla Voice

- Cortex-M4 x1
- Cortex-M0 x1
- Syntiant NPU x2



Upbeat UP201

- RISC-V SiFive x2
- NPU x2



視覺型開發板(MCU MVE+NPU)

ALIF Ensemble E7 AI/ML AppKit

- Cortex-M55 x2
- Ethos-U55 x2
- Cortex-A32 x2



新唐(Nuvoton) NuMaker-M55M1

- Cortex-M55 x1
- Ethos-U55 x1



Seeed Grove Vision AI v2 Kit (奇景 Himax WiseEye2 HX6538)

- Cortex-M55 x2
- Ethos-U55 x1



STM STM32-N6 STM32N6570-DK

- Cortex-M55 x1
- Neural-ART(NPU) x1



Infineon PSoC Edge Development Kit

- Cortex-M55 x1
- Ethos-U55 x1
- Cortex-M33 x1



NXP i.MX93 Evaluation Kit

- Cortex-A55 x2
- Cortex-M33 x1
- Ethos-U65 x1



Renesas EK-RA8M1 Evaluation Kit

- Cortex-M85 x1



Ambiq Apollo 510

- Cortex-M55 x1



台灣 MCU + NPU 開發板

奇景光電 (Himax) – HX6538

Seed Studio
Grove Vision AI V2



Seed Studio
SenseCAP Watcher



Grovetty
TinyTTS kit

WiseEye2 (WE2) Himax

- Arm Cortex-M55 2核 (150MHz, 400MHz)
- Arm MicroNPU Ethos-U55 (400MHz)
- SRAM 2432KB, 64KB Boot ROM, external 1~16MB QSPI Flash



<https://www.himax.com.tw/products/wiseeye-ai-sensing/wiseeye2-ai-processor/>

新唐 (Nuvoton) – Cortex-M55

NuEzAI-M55M1

NuMaker-X-M55M1D

M55M1/M5531

新唐科技



NuMaker-M5531



- Arm Cortex-M55 (220MHz)
- Arm MicroNPU Ethos-U55 (220MHz, 110GOPS) (only M55M1)
- SRAM 1.5MB, 2MB Flash
- 可外部擴充 HyperRAM/Flash, OSPI Flash

2M / 1.5M	M5531R2LJAE	M5531K2LJAE	M5531H2LJAE
2M / 1.5M	M55M1R2LJAE	M55M1K2LJAE	M55M1H2LJAE
Package	LQFP64 (10 × 10 mm)	LQFP128 (14 × 14 mm)	LQFP176 (24 × 24 mm)

<https://www.nuvoton.com/ai/product/>

瑞昱 (Realtek) – RTL8735B

RTL8735BDM
MODULE



Realtek
AMB82



Reltek
AMB82 Mini



Wistouch
HUB8735 Ultra



AmebaPRO2 REALTEK

- 32bit Arm V8M, up to 500MHz
- Intelligent Engine(NPU) @ 0.4TOPS
- 768K ROM, 512K RAM, 16MB Flash, DDR2/3L up to 128MB
- WiFi 2.4G/5G Hz, 802.11 a/b/g/n
- BLE 5.1
- up to 5MP or 1080P@45fps
- H.264/265 encoder

<https://www.amebaiot.com/zh/amebapro2-2/>

阿比特 (Upbeat) – RISC-V SoCs

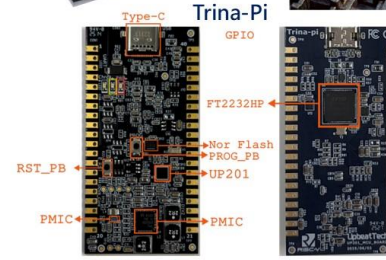
UP201



UP301



UP201/UP301 UpbeatTech



- 異質 RISC-V 2核 ·
Always-On(E21)@32K~11MHz
Non-Always-On(E34)@25M~400MHz
- 雙 NPU · NPU2 Always-On,
NPU1 Non-Always-On
- 可彈性調整計算效能
- SRAM 1.5MB, External OSPI
Flash up to 256MB
- GFX 2.5D GPU
- 8-bit Parallel Camera(only UP301)

<https://www.upbeattechtw.com/products/upbeat-soc>

MPU/MCU + DSP 開發板

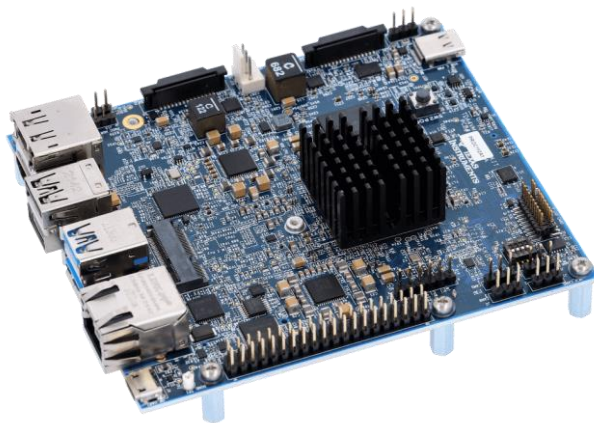
TI SK-TDA4VM

TDA4VM

Arm Cortex-A72 x2

8 TOPS AI

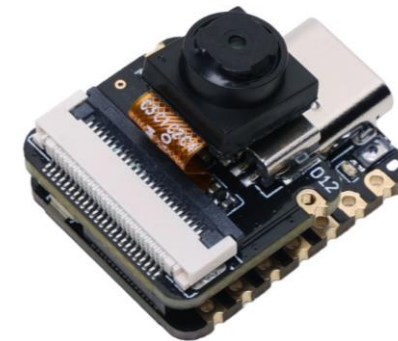
C7 DSP, GPU



Seeed Xiao

ESP32-S3 Sense

- Xtensa LX7 (DSP) x2



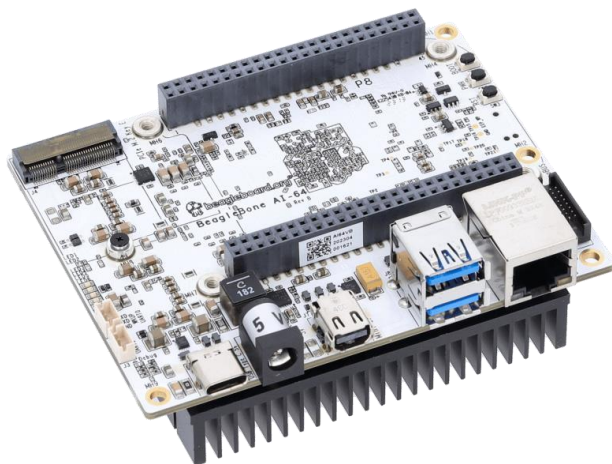
BEAGL-BONE-AI-64

TDA4VM

Arm Cortex-A72 x2

8 TOPS AI

C7 DSP, GPU



Seeed SenseCap Watcher

- ESP32-S3 x1
- Xtensa LX7 (DSP) x2
- Himax WE2 x 1
- arm Cortex-M5 x2
- arm Ethos-U55 x1



MPU + GPU 開發板

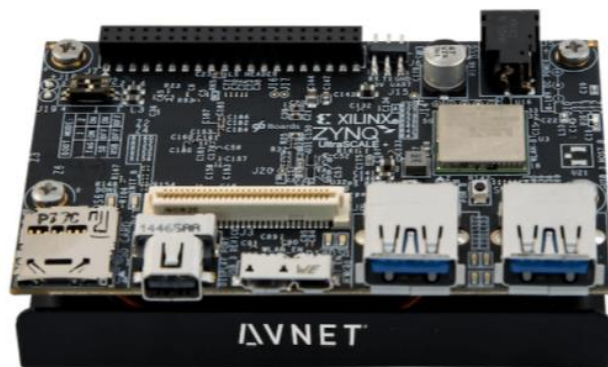
Nvidia Jetson Orin 家族 (Nano, NX, AGX)



	Jetson AGX Orin 開發套件	Jetson AGX Orin 64GB	Jetson AGX Orin Industrial	Jetson AGX Orin 32GB	Jetson Orin NX 16GB	Jetson Orin NX 8GB	Jetson Orin Nano Super 開發套件	Jetson Orin Nano 8GB	Jetson Orin Nano 4GB
人工智慧效能	275 TOPS		248 TOPs	200 TOPS	157 TOPS	117 TOPS	67 TOPS	67 TOPS	34 TOPS
GPU	2048-core NVIDIA Ampere architecture GPU with 64 Tensor Cores			1792-core NVIDIA Ampere c GPU with 56 Tensor Cores	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores		1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores		512-core NVIDIA Ampere architecture GPU with 16 Tensor Cores
GPU 最高頻率	1.3 GHz		1.2 GHz	930 MHz	1173MHz	1173MHz	1020MHz	1020MHz	1020MHz
CPU	12-core Arm® Cortex®-A78AE v8.2 64-bit CPU 3MB L2 + 6MB L3			8-core Arm® Cortex®-A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3	8-core Arm® Cortex®-A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3		
CPU 最高頻率	2.2 GHz		2.0 GHz	2.2 GHz	2 GHz		1.7 GHz	1.7 GHz	1.7 GHz
DL 加速器	2x NVDLA v2					1x NVDLA v2	-		
DLA 最高頻率	1.6 GHz		1.4 GHz		1.23 GHz		-		

資料來源：<https://www.nvidia.com/zh-tw/autonomous-machines/embedded-systems/jetson-orin/>

FPGA 開發板



AVNET FPGA開發板

Ultra96-V2

Xilinx Zynq UltraScale+
MPSoC ZU3EG A484

2GB RAM / 16 GB SD



AMD (原Xilinx)

Kria KV-260

Zynq™ UltraScale+™ MPSoC
系統邏輯單元數 256K

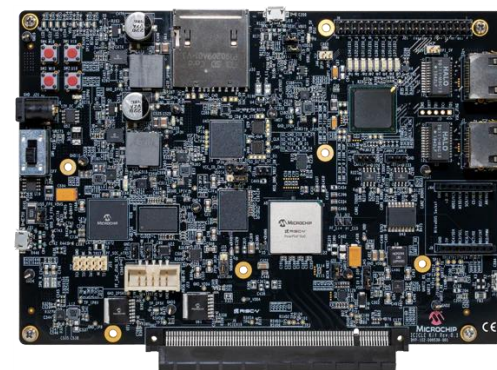
區塊 RAM / Ultra RAM數 144
/64



Microsemi FPGA開發板

M2S-HELLO-FPGA-KIT

SmartFusion 2 FPGA +
Arm Cortex –M3



Microchip PolarFire SoC Lcicle Kit

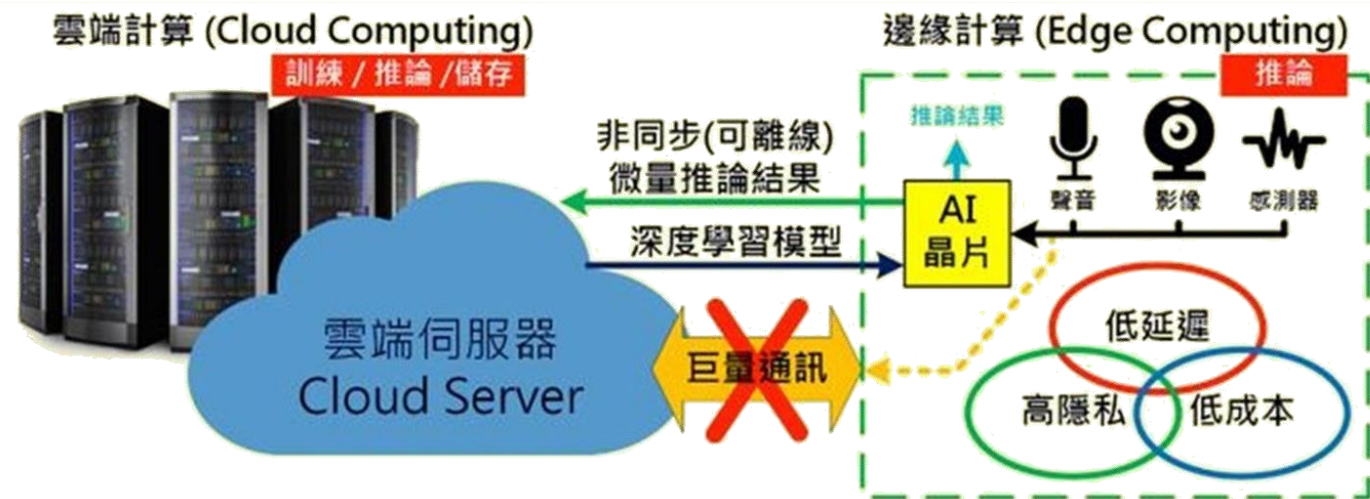
MPFS250T-FCVG484EES

系統邏輯單元數 254K

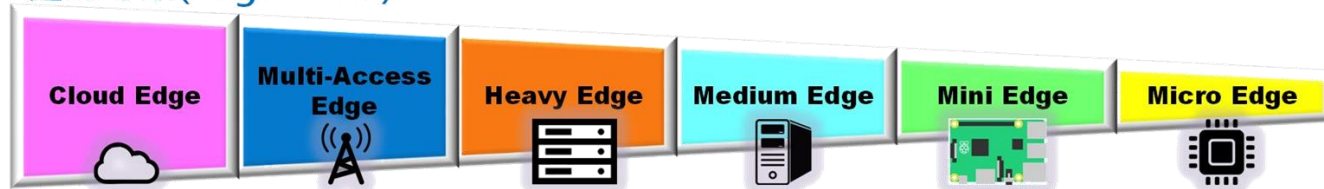
內建記憶體 17.6Mbit



Edge AI

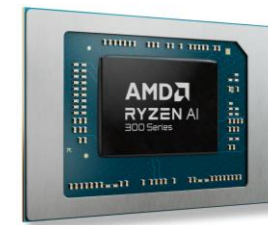


邊緣等級(Edge Level)



2.4. AI PC 級硬體

常見 AI PC 主要晶片比較



產品名稱	Intel Core Ultra 9 288V	Intel Core Ultra 7 165H	Qualcomm XIE-800-100	AMD Ryzen AI 9 HX 370
核心代號	Lunar Lake (V200系列)	Meteor Lake (V100系列)	Snapdragon X Elite	Strix Point
CPU核心	P-Core @5.1GHz *4 E-Core @3.7GHz *4 8 Cores, 8 Threads	P-Core @5.0GHz *6 E-Core @ 3.8GHz *8 LP E-Core @2.5GHz *2 16 Cores, 22 Threads	Oryon @3.4GHz *12 (Max. 4GHz) 12 Cores, 12 Threads	Zen 5 @5.1GHz *4 Zen 5c @3.3GHz *8 12 Cores, 24 Threads
消耗功率	17 ~ 37 W	28 ~ 115 W	23 ~ 80 W	15 ~ 54 W
內顯(GPU)	Xe2 (Arc 140V, 2.05GHz, 67 TOPS)	Xe-LPG (Arc Graphics, 2.3GHz, 18TOPS)	Adreno X1-85 @1.25GHz	Radeon 890M @2.9GHz
AI (NPU)	NPU 4 48 TOPS xPU AI Total 120TOPS	NPU 3 11 TOPS xPU AI Total 34 TOPS	Hexagon 45TOPS	XDNA 2 50 TOPS AI Total 80 TOPS

Apple AI PC 晶片

	Apple M3	Apple M4	Apple M5
CPU 核心數	4+4 4+4	4+6 4+4	4+6 3+6 (iPad Pro)
GPU 核心數	10	10	10 搭載神經網路加速器
RAM	24GB	頻寬120Gb/s 32GB	頻寬153Gb/s 32GB
神經網路引擎核心數	16	16	16



影像來源：<https://www.soft4fun.net/tech/news/m5-apple-silicon.htm>

Intel Core Ultra 演進



Meteor Lake

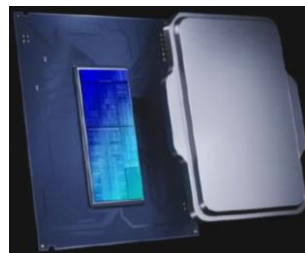
NPU 11.5 TOPS
iGPU 10~77 TOPS
Total 34 TOPS

(CPU+GPU+NPU)



Lunar Lake

NPU 40~48 TOPS
iGPU 53~67 TOPS
Total 120 TOPS



Arrow Lake

NPU 13 TOPS
iGPU 8~77 TOPS
Total 24~99 TOPS

Panther Lake

CPU 10 TOPS
NPU 50 TOPS
iGPU 120 TOPS
Total 180 TOPS

Nova Lake

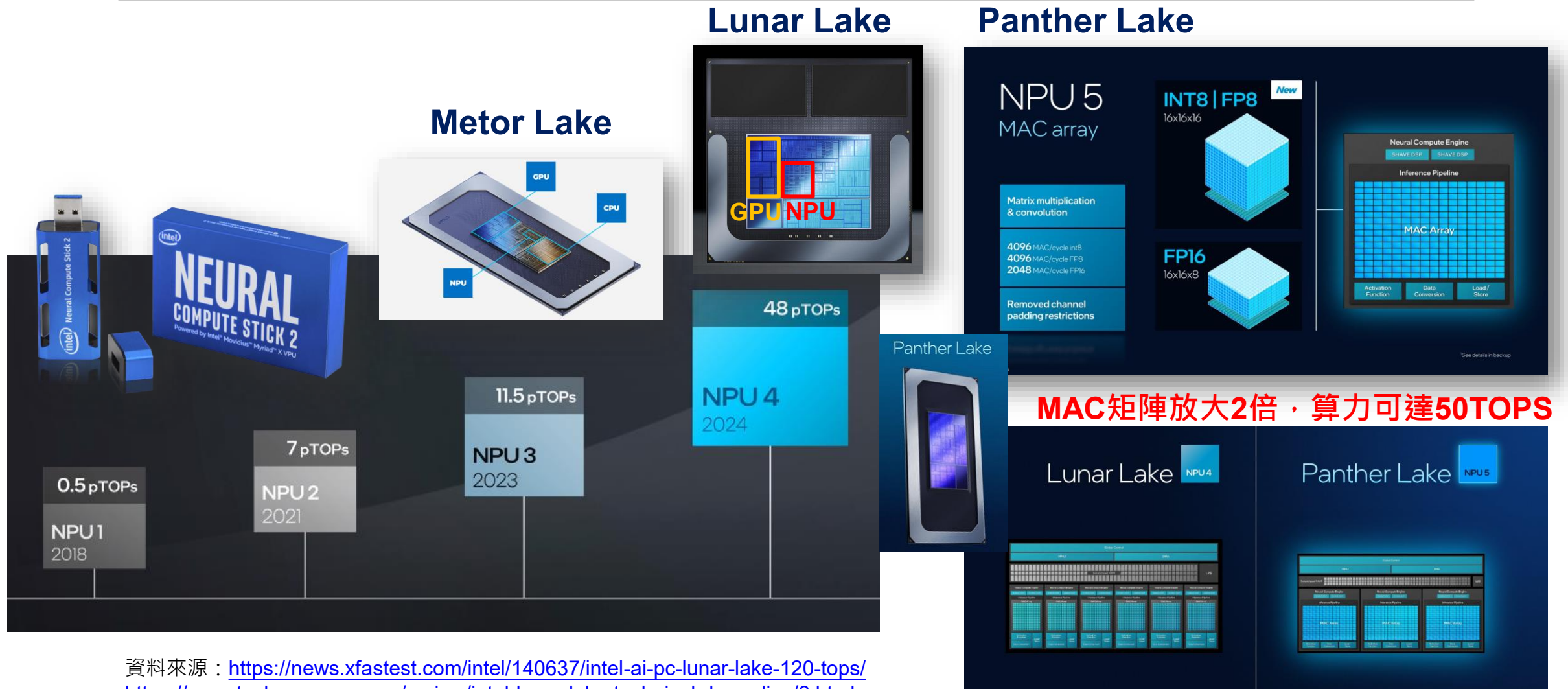
NPU 100+ TOPS
(預估)

Core Ultra 編號規則：



Meteor Lake 100 U/H 筆電用
Meteor Lake 100 HL/UL 嵌入式用
Lunar Lake 200 V 筆電用
Arrow Lake 200 K/KF/F/T 桌機用
Arrow Lake 200 H/HX/U 筆電用
Panther Lake 300系列
Nova Lake 400系列

Intel NPU 演進



資料來源：<https://news.xfastest.com/intel/140637/intel-ai-pc-lunar-lake-120-tops/>
<https://www.techpowerup.com/review/intel-lunar-lake-technical-deep-dive/6.html>

台灣生成式 AI 型硬體

聯發科 (MediaTek) – 5G 手機 SoC



天璣 9500
1x Arm C1-Ultra, 3x C1-Premium,
4x C1-Pro, Arm Mali-G1-Ultra
MC12 GPU, WiFi 7, BT6, NPU 990

天璣 Dimensity **MEDIATEK**

適合用於生成式 AI 及代理型 AI 推論



晶片型號	APU/NPU	AI 算力	主要用途
天璣 9000	APU590	約14TOPS	導入 ETHZ AI Benchmark v5 標準； 多模態融合運算
天璣 9200	APU690	約18TOPS	混合精度運算； AI 影像超分 (AI-SR) 優化
天璣 9300	APU790	約30~40TOPS	硬體級生成式 AI 引擎； 支援 33B 參數模型；LoRA Fusion 技術
天璣 9400	NPU890	約50TOPS	Agentic AI 引擎；50 tokens/sec 生成速度；時域張量加速
天璣 9500	NPU990	約100TOPS	雙 NPU 設計 (高功耗+低功耗)； 支援 BitNet 1.58-bit；4K 影像生成

<https://www.mediatek.com/zh-tw/dimensity-9500>

耐能 (Kneron) – GenAI Chip

KNEO300(EdgeGPT) 系列

Edge端伺服器，適用於生成式AI應用

- KNEO300 30TOPS(8x Cortex-A53)
- KNEO330 48TOPS(KL830)
- KNEO350 192TOPS



Kneron

KL1140

全球首款能在终端完整行
Mamba 模型架構的 NPU，
4顆級聯可運行1200億參數
之模型。



<https://www.kneron.com/tw/news/blog/251/>

創鑫 (Neuchips) – RecAccel N3000



RecAccel™ N3000
推薦推論晶片

neuchips



Viper Series
語言模型推論卡

- ARC HS48 CPU
- SRC EV72 DSP, FPU
- 64GB DRAM
- 矩陣運算引擎 × 10
(動態 MLP 引擎, DME)
- 向量運算引擎 × 2 (特徵交叉, FX)
- 嵌入式引擎



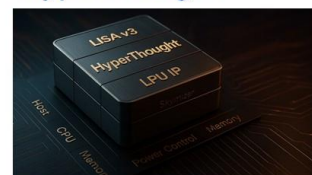
DM2 Series
語言模型推論卡

<https://zh-tw.neuchips.ai/products>

台灣發展軟體科技 (Skymizer) – LPU IP

skymizer

HyperThought



- 高吞吐量，100GB/S 只需
0.5TOPS 即可實現 30 token/sec
- 28nm 製程，多核可擴展設計
- 八核 HTX301 LPU 為 Llama 3
7B預填充實現240 tonken/sec

skymizer.ai/products/hyperthought/

EdgeThought



- 可將生成式AI應用引入各種邊緣
設備達到成本效益優化
- Skymizer 編譯器可優化執行效能
- 可使用自行研發高效語言指令架
構 LISA(v2, V3) 提高 LPU 效能

skymizer.ai/products/edgethought/

運算效能 — 主要指標

主要指標

- TOPS (每秒運算次數, 單位Tera, 10^{12})
- TOPS / W (每瓦運算次數)
- 算力成本 (每塊錢得到之TOPS)
- 數值精度 (INT4/8/16, FP4/8/16/32)
- 推論精確度 (Accuracy)
- 均值平均精確度(mean Average Precision, mAP)
- 推論速度 (Frame per Second, FPS)
- 批量/延遲 (Batch Size / Delay)
- 整體功耗 (Power Consumption, W)

大語言模型 (LLM) 指標

- 第一詞元(Token)產生時間
- 每秒產生多少詞元



運算效能 — ML COMMONS

AILuminate Benchmark

AlgoPerf: Training Algorithms

MLPerf Automotive

MLPerf Client

MLPerf Inference: Datacenter

MLPerf Inference: Edge

MLPerf Training: HPC

MLPerf Inference: Mobile

MLPerf Storage

MLPerf Inference: Tiny

MLPerf Training

Training



ML
● Commons

Inference: Edge

Area	Benchmark	Dataset	Quality Target	Reference Implementation Model
Vision	Image classification	ImageNet	75.90% classification	ResNet-50 v1.5
Vision	Object detection (light weight)	COCO	23.0% mAP	SSD
Vision	Object detection (heavy weight)	COCO	0.377 Box min AP and 0.339 Mask min AP	Mask R-CNN
Language	Translation (recurrent)	WMT English-German	24.0 Sacre BLEU	NMT
Language	Translation (non-recurrent)	WMT English-German	25.00 BLEU	Transformer
Language	NLP	Wikipedia 2020/01/01	0.712 Mask-LM accuracy	BERT
Commerce	Recommendation	1TB Click Logs	0.8025 AUC	DLRM
Research	Reinforcement learning	Go	50% win rate vs. checkpoint	Mini Go (based on Alpha Go paper)

Area	Task	Model	Dataset	QSL Size	Quality	Multi-stream latency constraint
Vision	Image classification	Resnet50-v1.5	ImageNet (224x224)	1024	99% of FP32 (76.46%)	50 ms
Vision	Object detection (large)	SSD-ResNet34	COCO (1200x1200)	64	99% of FP32 (0.20 mAP)	66 ms
Vision	Object detection (small)	SSD-MobileNets-v1	COCO (300x300)	256	99% of FP32 (0.22 mAP)	50 ms
Vision	Medical image segmentation	3D UNET	BraTS 2019 (224x224x160)	16	99% of FP32 and 99.9% of FP32 (0.85300 mean DICE score)	N/A
Speech	Speech-to-text	RNNT	Librispeech dev-clean (samples < 15 seconds)	2513	99% of FP32 (1 - WER, where WER=7.452253714852645%)	N/A
Language	Language processing	BERT	SQuAD v1.1 (max_seq_len=384)	10833	99% of FP32 (f1_score=90.874%)	N/A

資料來源：<https://mlcommons.org/>

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